



Going to outer space with new space: The rise and consequences of evolving public-private partnerships

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ARTICLE INFO

Original content: [Replication Data for: Going to Outer Space with New Space: The Rise and Consequences of Evolving Public-Private Partnerships \(Reference data\)](#)

Keywords:

Commercialization

New Space

International Competition

Status

ABSTRACT

What explains the commercialization of key government space projects through the incorporation of New Space? The newer generation of private companies have seen a significant increase in government contracting as they become instrumental for national security missions and high-profile civil projects. The turn to New Space companies, particularly those entrepreneurially-driven and privately funded, deviates from the governments' historical reliance on more traditional private partners. We argue that the turn towards New Space was neither inevitable nor monocausal, but rather the product of the confluence of the upstart sector's cost-efficient service offerings and rising public profile, which coincided with a period of renewed international competition. New Space firms indeed distinguished themselves by offering affordable products, an innovative production process, and a unique brand of prestigious reputation otherwise unavailable at national programs and older aerospace companies. However, these services were only deemed necessary for integration with the public sector because of the heightened importance of security and national status amidst a perceived return to great power competition. A new generation of public-private partnerships offered the only strategy for spacefaring states to attain and maintain a competitive position in an environment where non-state actors can match government accomplishments and capabilities. However, the utility of current integrative policy threatens to globalize not only the strengths but also the weaknesses of New Space, undercutting the very goals their adoption was meant to achieve.

1. Introduction

Why have governments chosen to incorporate commercial actors of the New Space generation into high profile projects and national security programs when such companies differ markedly from traditional private partners and introduce new risks?

In 2010, commentator Charles Krauthammer claimed that "It would be swell for private companies to take over launching astronauts. But they cannot do it. It's too expensive. It's too experimental" [1]. But within the decade, SpaceX's Crew Dragon-2 launch marked the first commercial crewed flight to the International Space Station (ISS) from U.S. soil. Notably, this was also the first spaceflight for NASA astronauts on a non-government launch vehicle. This achievement was the culmination of years of experimentation and heralded the advent of a new era of commercial space activities. Since this first launch, almost a dozen more Crew Dragons have brought NASA personnel and civilian astronauts alike to the ISS, and many more are scheduled under the auspices

of both NASA's Commercial Crew Program and private spaceflight [2]. The past decade has not seen such changes limited to the U.S. alone. New Space companies have increasingly become staple service providers internationally. From strengthening long-standing sectors like launch and satellite communications to becoming the partners of choice for future-oriented projects like lunar infrastructure, companies founded in the last few decades are becoming near-omnipresent [3,4].

However, the rapid growth of this vein of partnerships has raised concerns about the ability of governments to maintain control of their space programs. The worry that "private companies erode government's hold on space travel" [5] is analogous to concerns raised that other historically integrated firms have become less accountable to the national interest, advance profit motives first, and show less concern for international regulations [6]. Such lines of argumentation conclude that, especially for essential activities, governments should rely not on the private sector but rather government organizations for more tightly controlled and predictable agents. This debate undoubtably extends to

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<https://doi.org/10.1016/j.spacepol.2024.101626>

Received 11 February 2023; Received in revised form 9 March 2024; Accepted 16 March 2024

Available online 10 April 2024

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space, as notable disasters have shown that, especially for crewed spaceflight missions, modern space activities carry heavy consequences for which we would expect national governments to maintain the closest oversight.

Moreover, these endeavors represent the vanguard of technological progress, taken to reflect a country's values and scientific virtues. Spaceflight has long been presented as the "epitome of Western modernity" and served as a social standard used as a benchmark even beyond self-identified "Western" states [7,8]. To avoid exclusion from the recognized international community of innovative and technologically advanced states, a growing number of countries have initiated space programs as tangible symbols of their power, independence, and to reinforce techno-nationalistic sentiments. This close relationship to national status historically emphasized space as the focal point of acute competitions between spacefaring powers; the twentieth-century Cold War between the U.S. and Soviet Union and again in the twenty-first century for the U.S. and China. Throughout these periods, commercial actors have played a parallel role to governments, but one that has evolved significantly over time.

Contracting with private industries is not new for national space programs, military and intelligence organizations, and other government agencies, but the decision to diverge from legacy contractors and partner with less-established firms certainly is. The population of commercial associates was long limited to large (and increasingly consolidated) aerospace companies such as Boeing and Lockheed Martin in the U.S. and EADS/Airbus in Europe, and national space programs rarely deviated from these legacy contractors [9].

However, more modern recipients of contracts, such as SpaceX, Blue Origin, and Voyager Space buck this trend [10]. These companies are not descended from defense contractors of the Cold War-era, but rather emerged in the twenty-first century, and are funded primarily through private capital. Many such firms race to deliver products to market and tie their innovations to the potential for downstream services rather than waiting to respond to government-issued requirements [11,12]. Likewise, much of the emerging community can secure private investments and profitable market niches outside of government-oriented services. This camp, termed "New Space," is characterized by its entrepreneurial standpoint and a for-profit orientation that differentiates its members from their predecessors. Preceding generations have been retrospectively renamed "Old Space" [13] and remain defined by their close dependence on government-issued requirements [14].

New Space seemingly allows governments to enjoy the best of both worlds by shifting certain burdens to private firms while benefiting from their unique technological innovations. However, their inclusion carries important risks that make the increasing government reliance on their services distinctly puzzling. New Space, with its "fail fast, fail forward" outlook, differs markedly from traditional approaches to risk management in government R&D and procurement [15]. Thus, their participation can disrupt or delay projects if their contributions are less reliable, relative to offerings from their Old Space counterparts. Even well-funded investments have failed in recent years, sometimes only after consuming tens of millions of dollars without ever delivering their intended service [16]. Likewise, even realized projects, such as the Vulcan Centaur launch vehicle, may depend heavily on the delivery of New Space contributions, in this case Blue Origin's BE-4 engine, which was plagued by delays [17].

In parallel, profit motives and parochial interests may ultimately lead to misalignment of government national security and space exploration goals and the products and services that these firms prioritize. In the most high-profile instance to date, the intervention of SpaceX CEO Elon Musk to disrupt Starlink services in use by the Ukrainian military stand in stark contrast to the company's marketing of the telecommunications platform, and its militarized variant Starshield, as a viable service for the U.S. military to adopt [18]. SpaceX is slated to continue launching for the National Reconnaissance Office in the coming years [19], a key military service that, until 2017, was exclusively

the domain of Old Space company United Launch Alliance [20]. The potential for further cases like this makes the growing role of New Space in essential national security services all the more puzzling. The U.S. government's initial choice to incorporate New Space into these programs showcases a tolerance, increasingly also visible in other spacefaring states, to accept much greater ambiguity and vulnerability.

To explore this decision, we focus on major shifts at both the sub-national and international levels. This allows for an analytically eclectic application of international relations theory in the style of the "Space Policy Analysis" approach [21]. We draw on a theoretical framework used by scholars in international relations that utilizes the combination of opportunity and willingness to explain the proliferation of useful, but risky, technologies internationally [22–24]. The approach identifies the potential contributions and constraints/obstacles on the asset's offering side (the opportunity) and the incentives, concerns, or eagerness on the demand side (willingness). We argue that a combination of factors aligned to pressure governments into accepting both the risks and potential of New Space as necessary. In turn, the accomplishments of new public-private partnerships have driven the globalization of the phenomenon – and the risks it carries.

While the opportunity-willingness framework developed in the context of nuclear weapon acquisition and necessarily simplifies and condenses the numerous complex motives for techno-strategic policies, we now apply it in the new context of these public-private partnerships to emphasize the confluence of both categories [22]. The emergence of New Space actors, and their specialized innovations, was alone not a sufficient condition for increased government dependence for vital services. Rather, changes at the international system level brought urgent demand-side pressures, increasing willingness to partner with new private offerings in spite of the inherent challenges. Only the broader geopolitical shift towards competition amongst major spacefaring states, notably Russia, China, and the U.S., placed sufficient demands on policymakers necessary for the timing and rapid nature of New Space's integration.

On the opportunity side, changes in the commercial launch market drove the shift at the beginning, and set the basic pattern for what distinguishes the new generation of private actors. First, New Space companies swiftly design, test, and deploy highly capable platforms, most famously including reusable launch vehicles that governments like the U.S. had previously resisted. NASA's Space Launch System and traditional contractors involved in United Launch Alliance's Vulcan Centaur faced continued challenges despite over a decade of development apiece [25]. By contrast, examples like SpaceX's Falcon and Rocket Lab's Electron have become hallmarks of the modern market: more affordable while also offering quicker and more frequent launches. Second, many New Space companies avidly embraced alternative funding strategies. By spurning the continued dependence on government-first, commercial-second business approaches favored by Old Space giants, New Space companies eased the financial strain on governments [26]. Under alternatives like the fixed-cost contract, ballooning budgets and cost-plus agreements previously necessary to achieve a viable new product were no longer the only option. Third, the high-profile accomplishments of New Space companies offered a new source of prestige. Through public-private partnerships, governments could harness New Space to enhance their own domestic and international status.

On the willingness side, reliance on Russia for space launches became increasingly unsustainable for the U.S. and European governments amidst rising geopolitical tensions, especially after the 2014 invasion of Crimea [27]. Further, the reemergence of perceived great power rivalry restored space as an arena of heightened competition. Officials have adopted the rhetoric of overt competition, with NASA Administrator Bill Nelson formally asserting a space race between the U.S. and China [28]. Motivated by renewed tensions between foreign competitors and resultant concerns about the implications for international dominance, the U.S. redoubled its efforts to restore and expand its space access capabilities. Limited by stagnated budgets that increase

annually but form an ever-shrinking portion of government expenditures [29], willingness to integrate New Space services previously shunned rose accordingly both within and beyond traditional sectors like launch and satellite communications.

Our research makes several contributions. We offer an in-depth treatment of the rise of commercial space companies, which have received journalistic attention but less of a sustained scrutiny of their integration into national space missions, including the depth and breadth of that reliance [30–32]. Similarly, where recent initial explorations of New Space or the space agencies' move towards commercial enterprise focus on the international consequences [33–35] or models of commercial partnerships [36], we seek to address the reasoning behind the timing and nature of the continued shift. Despite the fact that neither New Space nor foreign policymakers anticipated their partnership, current government demand for new ways to access and utilize outer space provides a suitable market for New Space's innovations as a function of international relations as much as technology or economics.

Our research also speaks to the consequences of this increasingly global turn to New Space. We focus primarily on the U.S. case, as it is there that these developments first emerged. But the initial wave of commercialization there also catalyzed analogous efforts of foreign countries by providing a proven model for successful public-private partnerships. Although the specific format of partnerships varies widely across countries, most spacefaring states have sought to commercialize and outsource much the same services and integrate them with their own armed forces or terrestrial-facing services. Emulation drives the current proliferation of commercial space firms internationally, which will undoubtedly create new challenges for global governance.

We conclude that while the public-private relationship between New Space and governments has been fruitful, this integration introduces new vulnerabilities and challenges that will become only more salient. Because partnerships empower New Space companies financially and socially, they risk expanding their capacity to act in disharmony with state policy when private interests diverge from their partner's. In parallel, the growing number of non-state entities in space also raises serious questions about global governance. The speed at which commercial partners are integrated in particular reinforces concerns raised about the sustainability of increasing activities amidst increasing space debris [37], and the ambiguous position of companies as they embrace dual-use services and technologies. So long as these challenges do not receive uniform attention across states, there remains a serious risk to space as a peaceful, stable, and sustainable operating environment.

2. The U.S. Turn to new space

Reflecting the increasingly central role of commercial actors in space, the National Space Policy 2020 document frames its directive as to set out “the nation's commitment to leading in the responsible and constructive use of space, promoting a robust commercial space industry,” which it suggests is “foundational to economic development, continued progress, and sustained American leadership in space” [38]. The 2010 edition [39] similarly notes the need for a “robust and commercial space sector” but if repetition is any guide to priorities, the 2020 document references commercial industry at least twice as frequently as the earlier version. Under the heading of “promoting a robust commercial space industry,” the 2020 document notes that the U.S. should “Purchase and use United States commercial space capabilities and services, to the maximum practical extent under existing law.” The 2010 document has somewhat similar language but with more qualifiers, noting that the U.S. should seek commercial options “when such capabilities and services are available in the marketplace,” as they were far less mature in 2010 than a decade later.

In 2020, NASA celebrated the launch of American astronauts to the International Space Station on the first commercially manufactured and operated vehicle. For the previous decade, the U.S. had relied on Russian

rockets to access the ISS, and before that the Space Shuttle, which had since been retired. The U.S. could have continued its reliance on Russian Soyuz flights or the United Launch Alliance (ULA), a collaboration between Lockheed Martin and Boeing that touts its experience and successful launch record. Instead, the administration chose to rely on SpaceX, a newer company with more recent failed launches than ULA, but one that would go on to closely replicate their track record with a 99% success rate for its primary offering, the Falcon-9 launch vehicle [40].

Since the advent of the national space programs, private industry has been part of their discoveries, developments, and accomplishments. In the U.S., AT&T's Bell Telephone Laboratories first evaluated the feasibility of communications satellites in the 1950s, leading to a NASA test program, and in the 1960s they further developed infrastructure that became the basis of instantaneous global telecommunications [41]. Similarly, other private actors, primarily but not exclusively defense and aerospace companies [9], have contributed to NASA's entire project life cycle, bringing personnel, conducting research, manufacturing components, providing mission support, performing *ex post* analyses, and numerous other services. These companies are thus integrated into the policymaking and implementation cycle, with their reliability and capabilities already familiar to planners, who can therefore craft policy with their support in mind. In Europe, commercial actors were similarly present from the early stages, through participation in the Intelsat satellite communications consortium to the partnership with Arianespace for the European Space Agency's early launches [3].

However, the current range of private activities differ greatly from early cases. Where older efforts primarily took the form of well-established companies supporting government-led development and missions, the current landscape is markedly more speculative and ambitious. The participation of New Space firms characterizes involvement of the private sector in the twenty-first century. Many of these companies often resemble, or are, Silicon Valley start-ups. They are sometimes financed by individual entrepreneurs, use commercial off-the-shelf technologies, and adopt a more risk-acceptant and dynamic approach compared to legacy space contractors or state agencies [33]. Other entrants are established companies that hitherto offered only terrestrial services but have recently expanded into the commercial space sector, leveraging experience and resources from their established markets [12]. The advent of both types of New Space firms catalyzes research and development, with start-ups seeking to capture novel technological niches and demand-side firms opening up new applications for these offerings. Bridging these activities are “astropreneurs,” who connect space and terrestrial firms and customers with sponsorship, developing the networked aspects of commercial space and assisting with the harsh barriers to entry certain activities imply [42].

Reflecting the increasing diversity of opportunity and interest in commercial space, various Old and New firms have successfully managed to leverage vastly different approaches to secure necessary funding for the transition to the contemporary commercial environment. But where certain Old Space firms correctly identified government support as a viable augment to their robust offerings, they mistakenly assumed its continued necessity, leaving other firms to leverage alternative sources of investment [43]. What started with the creation of Blue Origin in 2000 and SpaceX in 2002 expanded over time, with 56 U.S. firms receiving investments by 2018, although 68% of venture capital funding between 2015 and 2020 went to SpaceX, Virgin Galactic, and OneWeb [44].

SpaceX has been the most successful and visible of the New Space companies, securing its first government contract from the U.S. Air Force and DARPA in 2005 and 2006, respectively, and its first civil contract from NASA in 2008. Its funding has increased through the 2023 fiscal year, where 72% of its government contracts come from NASA, 26% from the Air Force, and the remainder from an assortment of agencies [45]. Fig. 1 below maps SpaceX's proportion of total NASA obligations since 2005, showing a dramatic increase in SpaceX contracts from

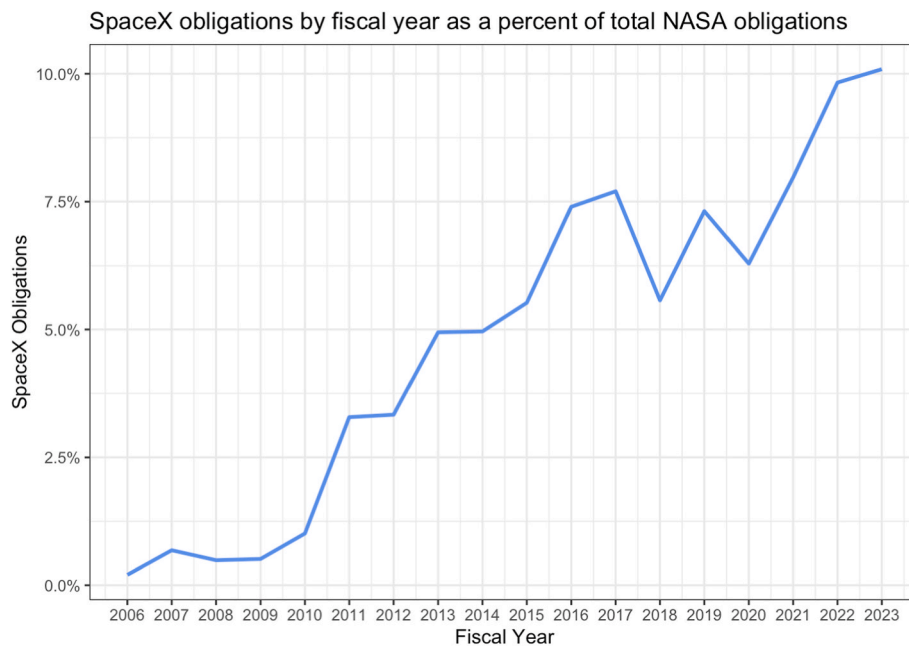


Fig. 1. Annual obligations (money paid to a contractor as a result of a contract award) paid out by NASA to SpaceX (y-axis) since FY05 (x-axis). Source: NASA Acquisition Internet Service [46].¹¹

near-zero to approximately 10% in 2023. This trend emerged amidst annual NASA obligations that remained largely flat, if not declining slightly for fifteen years, until a post-2020 spike that coincided with a dramatic increase in SpaceX's portion, as visible in Fig. 2. Effectively, SpaceX has captured a larger share of NASA contracts since 2005, while the annual value of all NASA contracts awarded has remained relatively constant. This dynamic is indicative of New Space's capacity to rapidly become integral to government activities.

3. Opportunity for partnerships and the willingness to pursue them

The willingness-opportunity framework stems from the work of Erik Gartzke, who suggests that scholars studying variation in international competition tend to focus on opportunities for conflict, relying on arguments that imply the default position of states is war [24]. Physical and resource constraints pose a challenge for all ambitious actors. But, Gartzke suggests that would-be competitors simultaneously also require an inherent willingness to fight, bear the heightened costs of resource-intense competition, and align the psychological incentives for decision-makers in order to overcome barriers to pursuing their objectives proactively.

When states share an affinity for common geopolitical objectives, the opportunity to instigate conflict, even opportunistically, is meaningless. This argument builds on earlier work by Siverson and Starr [47] that used national borders and alliances as markers to explain the probability of war diffusion, observing that these opportunity and willingness indicators are important in accounting for war "contagion." This process similarly holds for competitively-oriented technological developments, as Jo and Gartzke [22] and Singh and Way [48] attribute variation in nuclear proliferation across states to both opportunity (technological capabilities, fissile material, and economic capacity) and willingness (security incentives). In all of these contexts, the central theoretical assumption is not that states will automatically invest into geopolitical competition but rather that by default they refrain from policy changes unless they have both the potential to act (opportunity) *and* incentives, concerns, or eagerness that gives them a salient need (willingness).

The divide between opportunity and willingness is not clear-cut. In applying this framework, we recognize that these considerations are not

independent but argue that they provide a heuristic, simplified way of understanding a complex, symbiotic, and mutually-adaptive dynamic. We also acknowledge that these considerations do not mean that other compromises, needs, or motivations would not impact the decision of governments to work with New Space. As with development of nuclear programs, the process of space technology innovation is both risky and costly. The steep learning curve for developing new technological assets makes large-scale national space programs of the kind that could meaningfully invest into public-private partnerships a rarity. Bryan Early notes that only a handful of states achieve space launch capabilities [49]. While an even smaller pool of countries has access to a commercial space sector ready for public-private partnerships, we argue that many of the same logics should hold for explaining why a country reliant on government-driven requirements for innovation would then turn to New Space.

Although every country's path is distinct, we emphasize the case of the U.S. because of its historical leadership in space innovations. Because this position was previously achieved through reliance on Old Space companies, the decision to increasingly rely on New Space, and being the first to do so, sets this particular case apart as a test for the willingness and opportunity framework. For all other cases that follow, analysis must necessarily disentangle willingness and opportunity from the desire to replicate the first-mover advantages observed during the initial turn towards New Space. The U.S.'s willingness to reconfigure its policy undoubtedly influenced the decisions of other countries to attempt to do so as well [50].

4. New Space's opportunities

4.1. Efficiency and productivity

The opportunity side of the space commercialization story begins with the rise of New Space companies at the beginning of the 21st century. As *The Guardian* concluded in its analysis of commercialized space: "in actuality, there is SpaceX – and everyone else" [51]. Though New Space categorizes a generation of private space firms, SpaceX has indeed dominated, particularly in the last decade. In 2001, Elon Musk envisioned a "Mars Oasis," which would grow seeds on the red planet, but quickly realized the steep cost of space exploration, specifically for

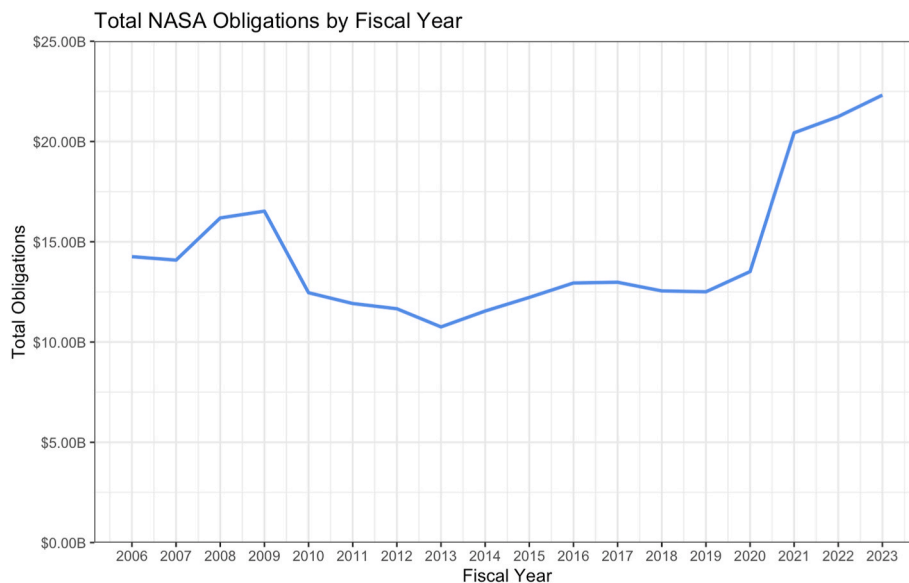


Fig. 2. Annual obligations (money paid to a contractor as a result of a contract award) paid out by NASA to contractors (y-axis) since FY05 (x-axis). Source: NASA Acquisition Internet Service [46].

the launch component [52]. Consultations reinforced expense concerns, as even if Musk spent millions of his own funds, launch costs would consume most of these resources. Needing a new approach to afford launches, he prioritized the development of smaller rockets and payloads as a cost-efficient alternative [53].

The purchase of Musk's prior company, Paypal, provided funds to overcome the high barriers to entry. In 2002, the Space Exploration Technologies Corporation, or SpaceX, was established in California, with an eye towards cheaper products and faster development. Their hiring process emphasized recruitment of engineers from top academic institutions, bolstering a reputation that attracted further talent in a self-reinforcing dynamic. A *Bloomberg* account suggests "Once the students looked Musk up on the Internet, selling them on SpaceX was easy. As word of SpaceX's ambitions spread, top engineers with a high tolerance for risk from Boeing, Lockheed Martin, and Orbital Sciences fled to the upstart, too" [54]. The Vice President of Human Resources at SpaceX suggests several traits that characterize a workplace culture which the company hoped would attract talent and set the conditions for a productive environment. An overarching ethos is that "we try not to limit our thinking except by the limits imposed by physics ... we build, test, break things, and iterate with a sense of urgency" [55]. This urgency is a function of company commitments to launches on short timelines, which places considerable demands on employees. However, talent was not deterred by the milestone-driven, high-stress culture, and indeed seems to have been attracted to it.

This work culture marked a stark contrast from NASA's public-side recruitment model. In 2006, NASA requested a study by the National Research Council to understand challenges in its workforce [56]. One key finding was that the space agency had little turnover, with NASA's tendency against hiring engineers directly from undergraduate programs pronouncing over time the effects of seniority, especially as the size of NASA's workforce diminished. At the beginning of 1966, NASA employed 420,000 staff, which by the end of the same year had already dropped to 396,000, and further declined to 307,000 in 1967, 265,000 in 1968, and 218,000 in 1969 [57]. By 2022, NASA had just 17,000 employees, of whom 1700 were civil servants [58]. Although contractors are more flexible, civil service positions are difficult to eliminate

and favor seniority, meaning that the labor force takes shape as a combination of attrition, retirements, and hiring freezes. The consequence is that, as the workforce study concluded: "NASA doesn't have a lot of people leaving, so what's been happening is that they're aging in place" [59].

In 2005, NASA hired just 411 engineers at a time when only a quarter of its engineers were under the age of 40, a proportion that would be about half by 2011. By 2018, more than half of NASA's workforce was 50 years old or more because of its low attrition rate, 4% annually, and propensity to stay past retirement [60]. Conversely, one member of the National Research Council suggested that "growing people straight out of college is kind of what industry does" [60]. With a workforce dominated by relatively young engineers, with the median age in the low thirties, SpaceX had access to staff trained at the cutting edge of their field. Their fresh experience pushed the envelope on new innovations, in a stark contrast to the workforce study's concern of path dependence associated with an older workforce.

This dynamic is not limited to SpaceX, but the company's experience offered lessons for both later New Space companies and government customers. Companies discovered that, in addition to growing private demand, government contracts were not limited to Old Space national champions, so long as newcomers could leverage unique experience to capture a specific technological niche. Their government counterparts determined that vertical integration was similarly unnecessary and embraced the value of overseeing a coalition of specialist partners. James Clay Moltz refers to this dynamic as a "netocracy," that produces a favorable division of labor between public and private researchers, manufacturers, and service-providers [61].

Assured by a favorable business environment financially and legally (for intellectual property) guaranteed by the government, New Space more freely embraced trends starting in the 1980s towards becoming a "composite sector." Therein, actors could focus on driving down the cost of a precise component within a broader service or product and still be assured of a viable market share on the other end by regulation and legislation [59]. U.S. federal and state governments had considerable room to facilitate the development of specialized actors, providing not only contracts, but also tax waivers, use of facilities and equipment, and legal protection on an as-needed basis for valuable partners [63]. The fruits of this labor are visible today, as New Space companies tout not the vast lists of products and services also found on Old Space websites, but almost uniformly emphasize their small, hyper-specialized

¹ Replication files for all figures can be found at <https://doi.org/10.7910/DVN/FQ2WTD>.

workforce and focus on delivery of much more specific offerings on an accelerated timeline. For governments still discovering their needs in the evolving orbital and political environments, the ability to pick and choose specialist partners who can deliver efficiently is attractive, as evidenced by the growing demand for “space-as-a-service” in all its various forms [64].

4.2. Alternative funding sources

A separate quality that reinforced the incentives for innovation within New Space was the operation on a different set of funding strategies. In the U.S., debates over government contract requirements paved the way for many of the largest New Space partners to make their case. NASA Administrator Bill Nelson termed the canonical “cost-plus” approach “a plague on us in the past” [65]. Under these terms, contractors received whatever resources and time necessary to deliver their product. Reliance on this format resulted in enormous cost overruns, and New Space distinguished themselves by embracing the alternative. “Fixed price” contracts, which only pay companies as they fulfill incremental project milestones, are now often the order of the day. This contract type incentivizes New Space companies to undertake far greater risks to succeed in the growing space economy, cultivating a market niche by offering viable services to government customers without the expense-creep risk of cost-plus. This “do-or-die” imposition is seen by some as key to the innovative accomplishments of New Space, particularly the development of affordable launch and satellite technologies [65]. Companies have gone bankrupt when unable to match the pace set by fixed-cost contracts, leaving those that remain as the apex of productivity, timeliness, and, in the best case, an ability to maintain that rhythm as customer needs evolve.

The new model for acquisitions upended the traditional policy of NASA. Previously, bespoke systems, where the government develops requirements, invites proposals from private bidders, considers alternatives among the responses, and awards cost-plus contracts served as the standard means of developing space capabilities [66]. For most established services, suppliers could perform risk assessments for true, revealed costs only once the project was underway. For novel projects, service providers might lack precise information on how volatile provision costs could become. What resulted was an undesirable reliance on cost-plus contracts to hedge against failure of the full investment by allowing for cost overruns.

Analysts suggest that some degree of reduced performance is increasingly acceptable if costs are sufficiently reduced [66]. Fixed cost contracts, and alternatives like Other Transactions Authorities (OTA) allowed for more predictable and stable budgeting. These alternatives reflect a confidence in service provision that is absent in truly “bespoke” services. They became plausible as New Space companies showcased an ability to provide viable services without severe cost overruns, thus mitigating the need for budgetary hedging. Reusable launch vehicles remain the quintessential example, as governments viewed the technology with skepticism and would have found cost-plus proposals risky. A newcomer to the market, without the established niche that guaranteed resources for long-term R&D, practically needed to embrace budget and schedule-efficient contracts to compete. And at the same time, the more the company was willing to bear the risks of the development cycle, the more their ability to deliver advanced capabilities offered a viable alternative to the bespoke model. Since the 2010s, reusable launch vehicles have increasingly become the mainstay of space launches, not only because of their technical capabilities, but because they set a precedent for the type of cost-efficient offerings government customers sought to invite.

An inescapable trend underlying that willingness resided in the budgetary constraints that the space program faced. Although orbital dominance garnered large shares of the federal budget during the Cold War, the post-1991 budgetary situation was constrained. Indeed, NASA’s budget has declined considerably since its peak in the 1960s

[31]. At its 1965 peak, the budget was \$34 billion (FY20 dollars) compared to \$23 billion in 2021, with a low of less than \$13 billion in 1975 and a post-Cold War low of \$19 billion in 2013. This pattern emerged as a compromise between the dual pressures on the space program, with supporters pleased with an annual increase in absolute funding, and budget hawks appeased by a decrease in relative share of the total government budget [29]. Reinforcing the trend are public preferences. Gallup polls consistently show that the U.S. public views NASA favorably. In 2021, 60% of U.S. respondents viewed NASA as either “excellent” or “good” and 83% feel that “the government’s space exploration at NASA” has “mostly been a good thing for society” [67]. Yet, only 27% of those respondents support spending more resources on NASA, compared to 66% who think spending should be the same or reduced and 6% who believe the program should end altogether [67].

In the meantime, several other trends emerged to compound the problem of stagnant, if not declining, budgets. First, the cost-per-launch increased [68] with exceptions for specific orbits [69], in part because of an acquisition model that consumed more of the flatlined contracting budget through cost overruns. Beyond the problem of cost-plus contracts, the U.S. had restructured R&D to ensure a reliable industrial base, but in so doing exacerbated the problem of inefficiency. In 1994, President Clinton signed a National Space Transportation Policy, giving responsibility for developing expendable launch vehicles with medium and heavy payloads to the Defense Department, which compiled specifications and contracted Lockheed Martin and Boeing to meet them [70]. Although there were two competing vendors, the contract guaranteed a minimum buy to the loser in order to maintain their business viability. Both companies recognized that even if they lost, they received 40% of the business, which led to high bids, high costs, and ultimately a merger into United Launch Alliance in 2006 [71]. Prices-per-payload for a ULA launch were roughly five times that of SpaceX, with 9 out of 10 missions facing sizable cost overruns [72].

In the meantime, the workhorse Space Shuttle program faced retirement, leaving a gap in the options for human space launch. NASA continually struggled to justify the high cost-per-launch, which precluded the project’s original goal of making space access routine [73–75]. As a result, the *Atlantis* flew the last mission in 2011, landing at Kennedy Space Center after 13 days in orbit and 30 years of Space Shuttle missions. With this transition came concerns of overcorrection in the name of budgeting. At the time, the Shuttle was the sole U.S. launch vehicle capable of carrying astronauts and had supported a broad range of satellite deployments, scientific projects, and even repair activities [76]. As early as 2008, analysts worried the Shuttle’s decline would mean a dearth of crewed travel to low earth orbit (L.E.O.) [77]. In 2010, Krauthammer noted with unease that, “by the end of this year, there will be no Shuttle, no U.S. manned space program, no way for us to get into space. We’re not talking about Mars or the moon here. We’re talking about low-Earth orbit, which the United States has dominated for nearly half a century and from which it is now retiring with nary a whimper” [1]. New Space’s more cost-effective contracting and alternative funding sources made its members especially attractive for surmounting these fiscal planning challenges, particularly as geopolitics pressured an (efficient) return to crewed spaceflight.

4.3. Status and prestige bootstraps

Status, referring to the perceptual hierarchy of states on the global stage, becomes an especially acute concern during tests of international orders. Status revolves around the positioning of actors within an informal and intersubjective ordering of the international system [78]. In this hierarchy, the actions and decisions of states are deeply influenced by a desire for status. This motivates active pursuit of opportunities to increase and reinforce their standing, especially against perceived encroachments [79,80]. Pursuit of status requires states to constantly gauge their reputation amongst peers, and shifts in this ordering serve as strong motives for interstate competition [81]. States

are concerned with status because of its many corollaries, as high status rewards with international recognition, cooperation, and even deference at the negotiating table with bargaining partners [82]. Scholars have also observed joint status-seeking by collections of states [83].

Previous scholarship on international relations and outer space have remarked on the combination of cooperation and rivalry there. For some scholars, space is a domain of prestige rivalry, but otherwise material cooperation (e.g., a space race, alongside a successful Outer Space Treaty [84]). For others, the commercial nature of space will have a pacifying effect on inter-state relations [85].

Scholarship on rising powers suggests that prestige competition is a mainstay. Rising states, especially, care not only for material increases in capabilities, but also recognition of these improved capacities by other states [86]. Status races are important not only for rising powers, but also more established status-quo states as well. Historical precedent places technological advancements like railroads as markers of civilization; and by extension, applies to how rising and established states consider space capabilities as similar symbols [7]. States pursue costly prestige projects, including space exploration, precisely because of how costly they appear [87]. Expending significant funds for seemingly intangible, nebulous, or distant rewards is a prerogative of wealthy and “advanced” states. Landing on the moon, after all, has yet to confer any strategic military advantage or vast store of material wealth, even some six decades later. States often show as much if not more concern for relative capabilities than absolute capabilities; and status scholars have, on the whole, made a compelling case for how relational perceptual order informs grand strategy.

^{OBJ}Conventional views of status-seeking space activities suggest that states close ranks around specific identities. In space policy, this dynamic appears in the notion of the “Space Club”, a phenomenon where states mutually recognize a restrictive definition of members who achieve advanced spaceflight capabilities [88]. Because clubs are exclusive in order to maximize club goods and reputational benefits, this framework originally suggests that the growing involvement of new, private players will erode the status of the club of states. We challenge this notion by positing that within the old Space Club, there is now a more exclusive *Commercial Space Club*. Instead of competing, public-private partnerships allow states to intentionally tie private successes to their own public image.

Bureaucracies are hardly immune to the dynamics of national status concerns. They seek to efficiently maximize the number of prestige-reinforcing activities and raise recognition of these actions. Bureaucrats might aim to harness the benefits of associating with the most prestigious firms, presenting their agencies as innovative and forward-thinking by means of a “Prestige Bootstrap.” The logic of this mechanism operates as follows:

First, advances in technology allow new firms to take part in specific aerospace enterprises for the first time and associate them with that progress. Both New Space start-ups (which introduce a technological niche) and experienced non-space companies (who carry over an established brand) may qualify for this role. Second, the reputation of these firms improves when they associate with national space programs. Partnerships serve to validate firms as capable partners, worthy of incorporation into missions of national importance. Their participation connotes trust on the part of the government, as well as recognition that the company represents the forefront of technological development, especially upon delivery of a successful joint product. Third, the national program can benefit not only materially, but also enjoy the prestige from collaborating with well-known private partners and efficiently producing high-profile products. A joint product allows the national program to sidestep accusations of “falling behind” private companies, and instead reframes them as a coalition that represents different aspects of the nation in union, rather than in separation or competition.

From these three premises, “bootstrapping” connotes a prestige-enhancing feedback loop. National programs regularly advertise the incorporation of technological innovators into their system of partners.

In doing so, they enhance their own status as leaders in space. Thus, policymakers can reward the winner of a contract by increasing their funding and reputation, as well as reflexively enhance their own status by virtue of partnering with them. As the status of the private partner increases after a successful collaboration, they can reap the benefits of their enhanced ranking by attracting additional investment and staff, further improving their value to the government customer. In sum, an initial injection of prestige and recognition through public-private partnership can bootstrap emergent New Space members into even more valuable future partners.

On its face, the notion that New Space would connote equivalent national-level power and prestige might seem dubious, but the achievements facilitated by its incorporation into state activities are key to heightening national status. A Pew Research survey demonstrates a strong foundation for New Space prestige, especially when paired with national civil programs [89]. A majority of Americans prefer for NASA to remain the primary actor, but a full third say that the accomplishments of private companies would be satisfactory on their own, and a strong plurality see the technology and accomplishments of private actors as laudable. Therefore, we may surmise that New Space has established itself as a notable brand that the public recognizes and judges positively. The significant portion of respondents who claimed private activities were preferable to the national program might pose a concern for the state’s attempt to accrue prestige. However, the survey never explores if respondents view partnerships between the civil program (which most favor) and the private programs (which most respect) as preferable to either working in isolation. It is that very middle path that the U.S. sought for its array of high-profile activities; partnerships with New Space members who hold the greatest value for national status.

Measures such as this bootstrapping approach are only truly necessary, however, when a state perceives a deficit in, or threat to, their relative status. Even states pleased with their status quo ranking pursue such measures if they view their favorable position to be in jeopardy by the aforementioned rise of an ambitious foreign power. Notably, high-tech projects that are human and capital-intensive and the attainment of recognized benchmarks of progress are very attractive to states seeking to establish a narrative of sustainable political ascent [90]. In turn, these accomplishments drive more established powers to deny rising states the would-be acceptance and recognition conferred by their own technological achievements [91].

One strategy towards this denial is to crowd out the competitor’s accomplishments, monopolizing the attention of the audience for status. Space technologies served this purpose during the space race, even when they offered few immediate material benefits [92]. Instead, they provided unique achievements for the superpowers to present as validations of their primacy [93]. The two Shuttle disasters were in a way a reputational nadir for the U.S. space program. However, the ability to leverage the rapid rise of private reusable launch vehicles has effectively resuscitated it. The *Atlantis* flew the last Shuttle mission in 2011, but as Erik Seedhouse states: “a year later [in 2012], following the successful flight of Dragon, SpaceX picked up the baton and is now carrying NASA’s legacy forward” [94]. Today, the Artemis lunar exploration program constitutes the primary attempt at status enhancement, presenting the endeavor as the first step towards future exploration. The concurrent Artemis Accords showcase an attempt at expanding soft power by demonstrating the breadth of global interest; implicitly in contrast to foreign coalitions like the planned Sino-Russian lunar base or proliferation of regional “space blocs” [95]. But for the Lunar Gateway project, corporate logos of private partners like SpaceX and Maxar are advertised on par with icons of state affiliates like Japan’s JAXA, Canada’s CSA, and Europe’s ESA [92]. Through such messaging, New Space is intrinsically tied to the brand of the Artemis program, and by extension, U.S. status.

Although opportunities for cost-efficiency and prestige arising from new actors in space were necessary for incorporation, they are not sufficient. Indeed, the case of Beal Aerospace is instructive. Founded and

funded by Andrew Beal in 1997, the company aimed to provide heavy-lift launch vehicles for space exploration. When the company shuttered in October 2000, Beal noted that “the most insurmountable risk is the desire of the U.S. government and NASA to subsidize competing launch systems ... unfortunately, development of a reliable low-cost system is simply not enough to ensure commercial viability, there will never be a private launch industry as long as NASA and the U.S. government choose and subsidize launch systems” [96]. As Beal implies, there also had to be a willingness to shift to emerging privately-funded firms, as we discuss in the next section.

5. The growing willingness to work with new space

Although the new capabilities provided an opportunity to diversify away from legacy contractors, governments also required a concurrent increase in demand for these services, ensuring the willingness to sustain the integration of New Space. Stagnant budgets and rising costs created a set of constraints in the investment environment in which the U.S. had to achieve its goals with the same amount of resources. Against this backdrop, two new geopolitical concerns made New Space’s technical strengths not only attractive from a cost-effectiveness standpoint, but necessary.

First, the end of the Space Shuttle program in 2011 highlighted a reliance on Russia for crewed launches; a concern that steadily grew as bilateral relations declined over the following decade. As the post-Cold War integration of joint projects became once more at odds with terrestrial geopolitics, exclusive Russian control over a key subsector of launch capabilities both threatened access to the International Space Station and called into question U.S. status. A purported leading space power could not maintain that position and simultaneously rely on a strategic competitor. Second, China’s rise and increasing activities in space allowed for it to once more become a venue for “great power competition” narratives. Legacy contractors were insufficient on their own to restore independent crewed launch capabilities and elevate U.S. status. These factors pressured the U.S. to overcome past hesitancy, embrace New Space firms as partners, and place them firmly within its geopolitical strategy.

5.1. Russia: from space collaborator to renewed space rival

As historian Mathew J. Von Bencke [97] observes, successful cooperation requires healthy bilateral relations, shared intentions for space policy, and greater utility compared to non-cooperative options. The collapse of these preconditions in the 21st century revealed an underlying concern about the structure of international space cooperation that required the incorporation of New Space to resolve. Complex interdependence, a condition in which intertwined economic relations modulates security concerns, is a major factor in inducing the recent privatization boom. The concept was espoused by “neoliberal” international relations scholars, who argued that international organizations and trade ties would compel states to act peacefully [98]. Interdependence research has a tendency to focus on trade and finance [99–101]. In relation to access to a particular domain, space, it is traditionally taken to be palliative, emphasizing international technological integration as a pressure in favor of cooperation and the dangers of non-compliance forcing restraint even amongst rivals [84,102].

By the late 1990s, however, scholars began to argue that complex interdependence could engender security concerns [101]; or that ties could be weaponized to give information or sanctioning advantages to certain actors [100]. Though once conceived as beneficial, interdependence was later cast in a more sinister light: bargaining leverage rose when partners tied together accrued asymmetric benefits from the interaction. The theory has thus far been used to explain great power supremacy in issues such as sanctioning, but oft less explored is how fear arising from weaponized interdependence could be a major factor in privatization booms.

Despite the long history of geopolitical competition amidst the development of national space programs, Russo-American collaboration was not inherently incompatible with concurrent political concerns. Beginning in 1975, as detente reduced tensions between the U.S. and the Soviet Union, both countries recognized the potential benefits of collaborating on space pursuits. The first major collaboration consisted of the Apollo-Soyuz mission in 1975, when a Soviet Soyuz 7K-TM spacecraft docked with an U.S. Apollo command and service module. Although the mission’s formal purpose was “to test the compatibility of rendezvous and docking systems and the possibility of an international space rescue” [103], the “handshake in space” between the two crews was a symbol for the “formal end of the space race” [104]. This single mission heralded the possibility of further Russian-U.S. cooperation, which reemerged after subsequent Cold War tensions. The inclusion of the Russian Federation in the development and construction of the International Space Station from the 1990s and later research activities conducted onboard exemplifies this. Indeed, the scientific enterprise of space exploration, even if competitive at times, nonetheless “lifted Soviet-American conflict out of the military sphere into peaceful competition, with astronauts fighting a symbolic battle, an alternative to nuclear war” [105].

This cooperation, however, always depended on political foundations. Building on the 1975 joint mission required first the activism of Soviet First Secretary Mikhail Gorbachev to reduce tensions and support a 1987 bilateral cooperation agreement for joint projects, and then the continued support of Russian President Boris Yeltsin to achieve the Goldin/Dailey-Koptev consensus on cooperation in the early 1990s [97]. Top-down pressure was necessary to overcome the fraught disagreements that emerged periodically, even amidst the proclaimed height of collaboration in the 1990s. Underlying budgetary and technology access motivations were never far behind political justifications. The subsequent rise of leaders with less conciliatory political motivations removed a key pillar for stable cooperation. Successive U.S. presidential administrations changed course, seemingly dependent on partisan affiliation. Whereas George W. Bush’s election ushered in a new techno-nationalist, unilateral stance on space security issues, the Obama administration saw a turn back towards international cooperation and outreach [106]. The oscillation reduced the predictability of bilateral relations, and set state interests at odds with prior trends towards internationalist commercialization [106].

Yet in Russia, Vladimir Putin’s succession after the conciliatory Yeltsin heralded a return to the pre-Gorbachev stance of ambiguous cooperation, a Russian strain of space-based techno-nationalist rhetoric, and full embrace of tight state-control of public and private space sectors [107]. However, the same neoliberal institutionalists who defined the possibility of complex interdependence in international cooperation note that, once established, links are difficult to detach, and even self-sustain without direct support to an extent [108]. With the observation that even states embroiled in total war will still trade in significant amounts with one another [109], it is not, therefore, surprising that the U.S. and Russia maintain some degree of cooperation, including continued partnership on the ISS. Thus, the removal of aligned leadership did not immediately end Russo-American collaboration, and has yet to do so entirely even today, but instead left room for wedge issues to motivate a search for greater freedom of action and alternatives to cooperation.

The break emerged as reliance on Russian equipment and logistics became a risky closing window for the U.S. and ESA member states. After the Space Shuttle program ended in 2011, NASA had turned to Russian Soyuz rockets to transport humans to and from the ISS [110]. But the reliance came at a cost: “NASA paid an average cost per seat of \$56.3 million for the 71 completed and planned missions from 2006” [111]. Likewise, Russian integration lent significant political leverage within negotiations on the ISS [112]. Critical dependence on Russian space assets extended beyond crewed missions and the issue of ISS access, with Russian components incorporated into core U.S. space

vehicles. Before the rise of New Space, private space companies were actually credited with filling in where intergovernmental ties faced challenges [97]. Old Space companies had the resources to persist through logistical challenges to cooperation, and saw significant efficiency gains for their efforts. But these collaborations left the U.S. Old Space companies as dependent on Russia as their government. In the clearest example, United Launch Alliance's Atlas V launch vehicle made use of Russian RD-180 rocket engines, with its ninety-nine launches as of 2023 serving as a testament to both the fruits of division of labor, as well as the vulnerabilities of overreliance.

Collaboration with Russia was thought to keep its engineers gainfully employed instead of lending their expertise to military missile programs, foreign rivals, and rogue states, as much as to preserve the ISS [113]. Nonetheless, even as the U.S. space program became increasingly reliant on Russia, "Russia being in the critical path" was always suspect because of unstable funding and quality control over manufacturing. Likewise, concern over Russian domestic politics and the tense relationships between Russia and other post-Soviet states closely followed all joint projects [97]. The existence of the Space Shuttle program had kept tensions in check, but its end greatly expanded the underlying risks of overdependence.

Souring political relations with Russia meant that reliance lacked political feasibility. Although cooperation could sometimes weather and compartmentalize away harsh political rhetoric [114], the influence of terrestrial security issues and increasing militarization of space challenged heuristic policymaking [115], presenting an increasingly inconsistent environment for joint efforts. In 2014, Russia invaded Crimea, prompting discussions over NASA's reliance on Russia for rockets, parts, and maintenance. In 2015, the National Defense Authorization Act prohibited the Defense Department from renewing or offering Evolved Expendable Launch Vehicle (EELV) contracts that involved Russian RD-180s [116]. However, since the ULA used RD-180s for their Atlas 5 launches, they amended the authorization and imposed a limit of 18 RD-180 engines that the ULA could order.

In response to this attempt to begin mitigating dependence, Russia retaliated. In May of 2014, Russia announced it would "deny the U.S. use of the International Space Station beyond 2020" [117], a claim it has maintained but not conclusively committed to through any formal policy [104]. Russia's Roscosmos, announced that it would not provide maintenance on the remaining RD-180 engines in the U.S. [118]. Collaboration with, and reliance on, Russia became politically toxic, creating pressure on the U.S. to revisit a domestic solution. As Administrator Charles Bolden argued, "The choice here is between fully funding the plan to bring space launches back to America or continuing to send millions of dollars to the Russians. It's that simple" [119].

Continued payments towards Roscosmos also became doubly problematic due to Russia's strong turn towards a competitive orientation and pursuit of space militarization, including outright weaponization. As Russia has declared its intention to partner with China for major projects, including a lunar station, concerns have emerged about technology transfers and the flow of resources to competing projects, with advocates for careful separation of the U.S. space program [120]. Russia increasingly prioritized security technologies, even as its civil program suffered from declining revenues [121], and revisited policies that endanger free use of space, particularly through testing of anti-satellite weapons [122]. The room for cooperation declined while the political and opportunity costs increased, and with them, willingness, if not need, to consider other options.

To decouple from Russia's space agency with regards to transport to and from the ISS, NASA had two options: develop a system "in-house" or seek partnerships with the private sector. Ultimately, they leaned towards the latter: "Bolden also repeatedly called for congressional approval of \$858 million in the proposed FY 2015 budget to support the development of U.S. commercial capability to transport crews to ISS, so that the United States was not reliant on Russian transportation" [119]. But as Old Space had invested heavily in links with Russian logistics, the

maturation of New Space companies without such dependencies made them the private partner by necessity. NASA's Crew Dragon Demo-2 launch in 2020 marked the end of its reliance on Russia and re-established U.S. self-sufficiency in human spaceflight.

The invasion of Ukraine in 2022 only reinforced the earlier decision to assert independence. Shortly after the beginning of the conflict, Russia formally cut off U.S. access to the RD-180 in a real test of the country's self-sufficiency [123]. By this time, however, the investment had paid off and officials were far less concerned. Secretary of the Air Force Frank Kendall claimed the U.S. should not be worried about meeting launch needs: "The whole point of the program we put in place several years ago was to work our way off of the RD-180" [123]. While Russia's space agency has indicated an intention to withdraw from the ISS [124], leaving NASA and its international partners, JAXA (Japan), ESA (Europe), and CSA (Canada) to maintain the ISS alone, this complication would not signal the death knell for crewed space flight or the ISS that it might have a decade ago.

Private sector firms have given NASA the capacity to transport crew and cargo to and from the ISS [125]; desirable for almost twenty years for cost-efficiency, but also increasingly necessary given the geopolitical pressures placed on space policy in the last decade. To illustrate the flexibility New Space has allowed, one can contrast the response to two threats made at two flashpoints of tension. In 2014, amidst disputes over sanctions imposed over the annexation of Crimea, Deputy Prime Minister Dmitry Rogozin stated "after analyzing the sanctions against our space industry, I suggest to the USA to bring their astronauts to the International Space Station using a trampoline" [126].

U.S. and European governments had little immediate answer, as they still depended on Russian launch services, limiting responses to self-criticism. But the response of New Space differed greatly, with SpaceX taking the opportunity to announce the Dragon 2 platform as an intended replacement for dependence. When geopolitical tensions led Dmitry Rogozin, then Director General of Roscosmos, to again taunt that "in this situation, we can no longer supply the U.S. with our rocket engines that are the best in the world. Let them fly on something else, like their brooms, or whatever, but at least we are freezing our shipments," [127], Dragon 2 was operational and several platforms were available or mature enough in their development to make the threat ring hollow. This turn began with, but has not remained limited to, the U.S., as European states sharing concerns over Russia's foreign policy have similarly turned to New Space to substitute vulnerable logistical links. When Russia blocked access to its facilities in retaliation against sanctions, New Space launch companies provided an alternative for the UK and its stake in the OneWeb satellite communications constellation [128].

"Decoupling" through domestic partnerships does not preclude useful U.S.-Russian cooperation in the future. The exit of Rogozin from Roscosmos has cooled rhetoric and crewed launches carrying personnel for U.S.-Russian joint missions have continued amidst current geopolitical disputes [129]. New Space has even integrated into the international relationship, with SpaceX providing launch services for astronauts from both states [130]. Decoupling has simply meant that cooperation can occur on mutually-agreed terms, without unfavorable leverage due to exclusive services or sunset agreements without domestic alternatives. As New Space provides consistent access to services even if governments are unwilling or unable, space activities can be insulated from geopolitical shocks and allow governments to preserve or even expand on collaborations where the lack of such options would mire space policy in fears of inequity and dependency.

5.2. China and the new space race

An important basis for a willingness to engage with New Space is the growing competition between the U.S. and China. The perception of space becoming an arena for "great power competition" placed twin pressures on the U.S. that New Space is suited to answer. First, as scholars have suggested, the emergent contest need not take the form of

overt military conflict, and there remains room for both cooperation and alternative, more stable, forms of competition [88,92]. If competition remains peaceful, confined to shows of prestige, New Space's contribution to status-seeking projects, and the ability of governments to accrue prestige through public-private partnerships is especially valuable. Second, if relations escalate to the point of conflict, New Space becomes urgently necessary due to the dual-use nature of its products and services. The possibility of either future gives governments sufficient reason to incorporate New Space, not only to take full advantage of its offerings but to hedge in favor of being prepared for whichever space environment they face.

China's emergence as a major economic power, closely in line with its entrance into the World Trade Organization, saw it subsequently pursue a series of high-profile accomplishments in space to demonstrate its technological capabilities and legitimize its own public image [131, 132]. China launched its first crewed mission in 2003, conducted its first anti-satellite weapon test in 2007, operated its original space station in 2011, and broke new scientific ground by exploring the far side of the moon in 2019. Although none of these achievements singularly constitute a "Sputnik moment," and indeed were framed first for domestic consumption, collectively they contributed to the perception of an emergent peer rapidly running the gamut of historical precedent to contest U.S. dominance of space. For example, the 2007 anti-satellite test, despite only replicating accomplishments already decades old for the U.S., was met with strong concern and treated as a breach of norms. The collaboration of NASA with ESA and the Russian space program to create the ISS allowed those parties to share in the prestige generated, rather than contesting it. China's exclusion from international cooperation with this group denied it access to a foremost area of the "Space Club," where they could have advanced technologically without as much suspicion. The resultant ambiguity has allowed some to frame China's accomplishments as inherently competitive, and denied the rising state the affirmative international recognition of its rising status that it seeks [80].

This dynamic has persisted despite consistent appeals to cooperation and rapprochement [91–93], perhaps in no small part due to the inherent security dilemma posed by the development of space technologies [94]. Because so many are dual use, they facilitate power projection, resource exploitation, and militarization, and any attempts to match and counter [95] those assets necessarily shifts the balance of power. U.S. policy analyzes stress separation of the two national space programs, particularly in terms of commercial space technologies [74]. The calculus of the U.S. government has also shifted sharply away from cooperation, as Congress in particular has coalesced around a strongly competitive perspective and enforced it through legal restrictions on cooperation [96].

Some observe that China's accomplishments do not imminently threaten the U.S., whose technological lead remains largely intact and prestige hardly eclipsed by a competitor's pursuit of feats already achieved in the 1960s. However, even if China has no intentions for status quo revision, its rise will be inherently destabilizing to the U.S.-favored Liberal International Order nonetheless, by nature of its illiberal institutions [133]. The combination of material resources and illiberal governance has automatically made China a major actor whose policy must be addressed. Across the selection of bilateral diplomatic interactions, there will inevitably be challenges and divergences of opinion for certain issue areas, such as the status of Taiwan. Because diplomacy often involves the bundling of separate issues, space policy inevitably links to these other points of contention and debate, and carries the potential to factor into crises over disputes.

In parallel to international pressures, domestic politics is no less responsible for the deterioration of bilateral relations and the consolidation of "great power rivalry" narratives [134,135]. On China's side, the centralized structure of its institutions makes it difficult to signal true intentions transparently to the U.S. [136]. Reinforced by the consolidated nature of China's space program, without civil or military

distinctions, and its commercial space sector of largely state-owned enterprises, U.S. policymakers may find it difficult to distinguish feats for domestic consumption from long-term strategy [131,132]. Interest groups that utilize populist and nationalist rhetoric can instigate a tense rivalry where material shifts are necessary but not sufficient on their own [137].

The atmosphere of ambiguity and open-ended questions about the future of the international system has empowered more hardline policy entrepreneurs and sidelined figures more inclined to accommodate. On the U.S. side, the maturation of the New Space launch sector and the close of the Shuttle program also coincided with the ascension of a broad cohort of hardline legislators whose public rhetoric matched their policy stances [138]. The rise of these figures also pressured incumbents labeled as too dovish to harden their stances to fit the developing consensus. Because hardline stances on one side reinforce hyper-nationalism and promote hawkish policymakers on the other, the "means" of seeking and maintaining a venerable status in space easily become "ends" in and of themselves [135]. As such, even if the lack of imminent challenge to U.S. hegemony is correct, some form of competition has become inevitable as a function of domestic politics in the U.S. and China. Therein, New Space companies play a vital role, even as their home countries endeavor to maintain separation, in keeping that competition as peaceful as possible.

U.S. officials have long embraced the narrative that no domain is more high profile than space. But, in pursuit of status-enhancing prestige from space, status competition has been, until recently, state-exclusive. During the Cold War, the last instance of acute status competition in space, advancements signified the superiority of one's own national political system vis-a-vis the other. Fearing that its global image as a scientific power fell behind after the Soviet launch of *Sputnik*, the U.S. government made huge investments in space exploration. And when Apollo 11 landed the first Astronauts to the moon in 1969, the Eagle lander vehicle was emblazoned with national symbols of "USA" and "NASA"; overt national symbols, and taken to signify the apex of the status contest between the superpowers, at least in space.

As China's status rose with its accomplishments, its presence outside the U.S.-led order therefore invoked a challenge. Russia's aforementioned turn away from collaboration, essentially a voluntary exit from the post-Cold War order, only reinforced that threat. The danger worsened with the rise of New Space, as the end of the Space Shuttle program, contrasted with the subsequent rise of private reusable launch vehicles, threatened to reinforce the notion of declining capability. SpaceX's Falcon-9 launched for the first time in the same year that the Obama Administration announced the cancellation of Constellation, the planned program to return to the moon and a would-be successor to the Shuttle. Inaction would have seen U.S. status continue to decline, ceding ground to rising foreign and private actors.

The U.S. struggled to thread the needle between a renewed interest in prestige projects to reinforce its flagging status and the concurrent obstacles to its ability to pursue this objective. However, New Space did not exclusively pose a threat, but could instead serve as a key asset in augmenting renewed government attempts to utilize space as a prestige-generating arena, not as a function of their utility but out of necessity. To reinforce national status, the emerging competitiveness has put pressure on the U.S. to return to space, achieve new accomplishments, and do so often. Due to continued budget restrictions [29], reliance on legacy space industry was too limiting, given the cost of ULA launches.

Turning to New Space for more launches allowed the U.S. to optimize on a number of dimensions. New Space is more affordable than the legacy contractors and therefore helps conserve resources, but not at the expense of Old Space. Moving away from a sanctioned monopoly allowed the U.S. to maintain two independent paths to launch its security and prestige-seeking missions. The government pays ULA \$1 billion a year to maintain Delta, Atlas, and Vulcan vehicles to essentially keep on retainer the engineering talent at these companies [139]. This redundancy, normally a mainstay of strategic systems, was borne by the

competitive pressures of the international environment. The more efficient the budgets, the more missions NASA could afford to conduct, and therefore maximize its contribution to national status.

Then, the vast increase in space launches by private companies, carrying corporate logos, but publicized on their own social media as in support of national security and scientific objectives, typifies the “bootstrapping” method we’ve outlined. If the U.S. could not be seen as ceding space to private companies alone, partnerships tie the success of the victors to its own prestige. Private entities are increasingly providing capacities to state ambitions in the domain, either directly via the provision of specific solutions (such as launch capacities) or indirectly by running important services (such as global internet coverage or photo reconnaissance data) that can be regulated by a state in which the company resides.

With the policy set to develop space-based capabilities in the national interest, then the logical approach is to break the 60-year mindset of building bespoke space capabilities and introduce the most innovative options possible, which today are often commercial [45]. Space launches are globally-viewed events, placing national accomplishments on the international stage. Just as the Apollo program in the 1970s boosted U.S. credibility and soft power [140], the country that can attain major accomplishments can make claims to global leadership regardless of whether those capabilities are developed by the Jet Propulsion Laboratory or SpaceX. With more ambitious outcomes, the space program can again claim pioneer status in the context of a geopolitical race with China that has, as in the Cold War, viewed discovery as one of the measures of performance [141].

The perception that some prominent New Space companies are in actuality extensions of U.S. interests is a consequence of this design. New Space is indeed tightly integrated into U.S. endeavors, and maintaining the sector’s subordination to the government preserves its position as a contributor, not a competitor. By investing in New Space companies in their infancy, the state can ensure they align with the national agenda, and therefore helps guide their activities towards providing essential services. Space competition “is tied to the cooperation with private entities, directing their activities to aid the national strategy of a given state” [35].

There is, however, always the risk that status competition will be insufficient to satisfy the U.S. and China, or that geopolitical tensions may contribute to the outbreak of genuine conflict. Why might we expect such “adversarial” pressure rather than cooperation? The space race paradoxically ended with peaceful relations in space and the Outer Space Treaty of 1967 was a watershed moment of cooperation despite the ongoing Cold War. Whether similar cooperative mechanisms will arise with China is yet to be seen; though the U.S. still maintains the most space assets, China’s rise may require additional cooperative overtures to facilitate peaceful use of space [142,143]. If domestic coalitions are unwilling to actively support reassurance, however, then the space race may become a less appropriate comparison. Cooperation versus conflict is, fortuitously, not unidimensional. China is indeed cooperative across certain forums; yet adversarial across others [144]. Indeed, it is certainly plausible that, because of the dramatic economic usefulness of space systems, there will be a cooperative atmosphere in outer space that precludes military conflict [85].

Yet we still have reason for caution. Unlike during the Cold War, space is now inordinately congested and the advent of reusable launch vehicles implies that available orbital “real estate” will rapidly diminish in the future. Economic gains that China receives from space, with regard to relational power vis-a-vis the U.S., must therefore have a hard ceiling. Even before they approach this ceiling, terrestrial actors will have to tread carefully, decrease deployment of space assets, and invest in active debris removal to reduce the risk of Kessler Syndrome [145]. Even real estate on the re-emerging targets for exploration of the moon is not created equal, with the Peaks of Eternal Light being more useful than other locations [146]. Cooperation in the prestige-based style of the space race may not satisfy states because space is asymmetrically more

useful to first-movers than newcomers. Consequently, we surmise that the zero-sum nature of L.E.O. may thus be the bane of simple cooperation.

Fears of military confrontation, should relations deteriorate past status competition, and the vulnerability of critical satellite assets to recent Chinese technological advances [98,99], have pressured the U.S. to prepare to replace destroyed satellites quickly during a crisis [100]. Adding additional launch providers, particularly those with reusable capabilities, and operators of private services compatible with military infrastructure would be critical towards this goal.

The situation arising in space is not dissimilar to that in the artificial intelligence “race”. Because artificial intelligence has considerable dual-use implications, private actors from a civil background found themselves integrated into national security projects [147]. Overall, the dual-use nature of nearly all New Space offerings have already prompted foreign states to declare private actors fair targets in an inter-state confrontation [148–150]. Therefore, if refraining from integrating New Space may not save them from being attacked as if they were, then governments have little reason to hold back from public-private partnerships. Furthermore, full integration may be the key to military deterrence, just as increasing dependence on space infrastructure already serves as an economic deterrent [85]. If omnipresent private infrastructure makes decisive first strikes against military assets less valuable, states will have much less reason to fear outright conflict. For these reasons, New Space is seen as both a critical component of peaceful status competition, a bulwark against actual kinetic clashes, and an inescapable factor in the event of such conflict.

6. Alternative explanations

While we argue that New Space’s features made the community especially productive, simple cost-effectiveness cannot explain the speed and dramatic magnitude of the subsector’s incorporation into the top tier of public-side projects. Even though major members of New Space, such as SpaceX and Blue Origin [151,152], have engaged in lawsuits, with variable success, to press for access in contracting, once they gain entrance their integration into major projects is rapid and substantial. Given that technological superiority must be, to some degree, incubated and adapted to fit government applications, not merely found, why was the “modernization” of Old Space not simply prioritized instead? Indeed, the choice to accept, and promote, New Space in its semi-independent format as essential partners instead showcases the strong desire of policymakers to access their services quickly, rather than prioritizing the stability of their integration.

A second alternative explanation is that the U.S. government is ideologically laissez-faire, and thus has accepted New Space on the grounds of promoting the free market. Indeed, more neoliberal-leaning administrations pushed for privatization starting in the mid-1980s, continued attempts into the early 1990s, and then legislated towards deregulation with renewed fervor during the early 2000s. One may contend that the modern commercial space boom was inevitable due to legislation such as the 2004 Commercial Space Launch Amendment Acts at the latest, or the 1984 Commercial Space Launch Act at the earliest. However, we must also account for the willingness to support “national champions” (large state-backed enterprises) instead of embracing New Space wholesale. After all, regardless of executive-side ambitions, congressional buy-in with New Space has been fraught with suspicion and caution from the beginning.

Only after significant negotiation and legal battles has Congress become more supportive of New Space’s incorporation into the routine community of NASA and defense contractors. Likewise, a 2016 SpaceX Falcon 9 explosion immediately garnered concern from a chorus of officials over safety, and the grounding of Starship flights after the April 2023 testing failure shows continuity in such concerns [153]. A government seeking laissez-faire commercialization would not have required one of its favored partners to pause their testing until stringent

government concerns are met. Instead, both the U.S. and foreign governments maintain their support of legacy Old Space companies, and have only begrudgingly embraced the diversification of the market as a function of geopolitical necessity, rather than full-throated support.

A final alternative explanation has been posited by economic historian Alexander MacDonald, who argues in his work, “The Long Space Age,” that commercial interests have always been at the root of space exploration [154]. However, MacDonald contends that government dominance of space activities peaks during periods of geopolitical conflict (i.e., the Cold War) [160]. However, this conceptual framework would suggest that we should be experiencing an increase in public, rather than private, dominance of space activities due to geopolitical competition. MacDonald argues that U.S. leadership seized the reins of space exploration to manage a signaling game, embodied in the Space Race [160]. But while high-prestige projects have made a reappearance in U.S. space policy, we observe what appears to be precisely the opposite phenomena vis-a-vis the American and European response to the rise of China. Instead of asserting the role of public spending and government leadership, states are *expanding* the aperture of private actors who provide essential space services. Public funding for NASA has not dramatically increased despite the semi-acceptance of the “new space race” framing. Our argument suggests that geopolitical rivalry engenders a special receptivity to New Space, and therefore contributes to the “Long Space Age” in a manner rarely seen historically.

7. New space, new problems?

Despite the demonstrable value of New Space’s capabilities, public-private partnerships create not only new opportunities but also new risks. The flow of government funding to New Space, on top of market growth, has greatly empowered these actors. However, their participation in public-private partnerships does not guarantee an alignment of public and private interests, leaving much room for divergences that we have already begun to observe. These divergences become especially problematic as New Space’s integration blurs the boundaries between public and private sectors, posing a challenge for international space law and global governance. The globalization of New Space partnerships risks vastly expanding these concerns internationally. New Space is both too useful and necessary for states to simply avoid utilizing. Thus, each challenge requires concerted attention, domestically and internationally, to limit the inescapable costs of incorporating New Space.

7.1. Divergent public and private motives

New Space firms may pursue the ambitions of their idealistic founders rather than prioritizing broader terrestrial services. Major New Space companies famously began as passion projects, with goals like making humanity a multiplanetary species, which is not an explicit goal of national space programs. For example, Blue Origin’s stated objective is to provide a future in space that millions of people can call their office and home, while SpaceX explicitly seeks to facilitate the colonization of Mars, despite the legality of colonization in space remaining an open question for governments. Even if temporarily aligned with public objectives, the value of, and sheer resources invested into, New Space has also provided significant political, social, and financial capital and may also empower companies to act on much more serious divergences from public interests.

The entrepreneurial leaders of New Space shape their companies’ behavior. Unlike Old Space, the centralization of some New Space companies around founders transforms their leaders into single points of failure. Whereas Airbus or Lockheed Martin maintain hundreds of millions of shares floating on an open market [155], privately owned SpaceX’s voting shares are held predominantly by Elon Musk himself [156]. Many Old Space companies may still bear the names of their founders, but these past generations largely began their forays into space as contributors to government requirement-driven projects, and have

not deviated from this supportive position. New Space leaders often began their ventures with lofty goals not necessarily grounded in practicality and certainly not in an immediate national security imperative. However, their insertion into major government policy has, for the first time, empowered them to influence, intervene in, and interrupt these policies.

The political inclinations of entrepreneurs highlight concerns about their reliability, as founders insert themselves into sensitive political debates and take stances that may deviate heavily from that of their ostensible government partner [157]. Starlink satellite access was withheld from supporting a Ukrainian attack on Russian-held Crimea in September of 2023, for instance [158]. “Astro-Libertarian” founders have made gains amidst the New Space wave, and in turn support many of the most risky endeavors [159] or promote deregulation of (inter) national space law that favors their freedom of action but enables those frivolous or dangerous projects. Their activities apart from government projects have already expanded the number of adherents of these figures, to the point where individual entrepreneurs achieve an almost messianic public appeal [160]. This dynamic is especially problematic because the increased stature of these entrepreneurs and the scale of their following risks legitimizing attempts by them to influence international norms, national policy, and the trajectory of space activities in general.

The incorporation of New Space into government projects is in no small part responsible for this appeal, which has been borne out heavily by prestige bootstrapping. Public-private partnerships allow private companies to share in the prestige of the common endeavor. In doing so, they have elevated the brand of astro-libertarian entrepreneurs, and further provided them the legitimating shield of government association. Because prestige is a psycho-social conception, its use is not statically materialistic in increasing profits and lowering budgetary costs only. Rather, it is folded into the broader myth-making narratives of an exceptionalist frontier conquest [161], a national calling, and the achievement of space-faring utopia [160]. Thus, the increased stature of New Space inherently risks endorsing its “market-oriented” vision for space policy over other “socio-technical imaginaries” [162]. This catalyzation of the narrative of democratized access and economic liberalization surrounding, and promoted, by New Space has immediate consequences for the practical conduct of regulation and diplomacy. These imaginaries possess important political, economic, normative, and environmental implications that threaten equitable and sustainable access to space. The rhetoric around “New Space” is inherently elite-led and economically oligarchic. Sustainability concerns, especially of space debris, are not seriously incorporated within this discourse; nor is space considered the “shared” domain of humanity once considerations of economic utilization enter discussions [162].

Old Space has achieved recognizable brands, and already shows potential to influence space policy through lobbying, but never has the same celebrity public appeal ascribed to New Space. The simultaneous financial and social empowerment of New Space carries the potential to magnify bias and myopic influencing of the public agenda. New Space actors maintain tremendous power within a band that is meant to be the shared domain of all humanity, and ceding control over either the pace and direction of space endeavors greatly increases the dangers for humanity as well.

Regardless of whether a company is beholden to an individual figure, New Space companies can also provide room for investors and foreign interests to insert themselves, which raises further concerns for bias and the security of government projects. The entrance of sovereign wealth funds into the field of investors suggests that New Space actors may find their interests increasingly transnational, and thus further complicate the selection of objectives and alignment with government needs. For example, in 2022, SpaceX announced that it would be raising a new round of private equity funding to pay for two capital intensive projects, the global satellite internetwork Starlink and the next-generation rocket Starship [163]. These investors include not only venture capital but also

national sovereign wealth funds, implying that a large set of private and public actors seek access as a commercial partner rises in importance to their government. SpaceX has not released the exact proportion of its government versus private revenues, but NASA and military contracts are thought to account for a high proportion of SpaceX's revenue [164].

Predicating New Space public-private partnerships on integrative relationships that offer continued government business to New Space partners may maintain incentives for companies to continue anticipating and providing government needs. The possibility of using emergency measures if the supply of main contractors is inadequate is ever available [165]. However, if the success of major public-private partnerships becomes self-reinforcing, it could erode the diversity of the industrial base by facilitating quasi-monopolies.

Such market dominance would make the commercial space ecosystem less competitive over the long term, much as Old Space contractors had become after years of government largesse. But becoming an indispensable national champion may also empower New Space actors to bypass limits set by their governments. In a recent advisory comment of Federal Aviation Administration's application of the 2015 Space Launch competitiveness Act, SpaceX prefaces its recommendation for avoiding new regulations and reducing prescriptive legal requirements with a reminder that the company is both a contractor through at least 2030 and the sole U.S. provider of crewed spaceflight [166], demonstrating the company's ambitions towards determining its own regulatory operating space. Likewise, a New Space actor that establishes a hold on limited orbital operating space, as appears to be occurring with the vast growth of private L.E.O. communications constellations, cannot simply be dismissed in favor of an alternative contractor. The greater the dependence on New Space companies for technological, infrastructural, and service niches, the more governments reinforce the risk of costly divergences from the public interest.

7.2. Challenges for global, and orbital, governance

The integration of New Space actors into government contracting may lead to greater complications of space governance. The growth of commercial firms engaged in space-based services, operating commercial satellites, and seeking government contracts has increased considerably in the last two decades, raising questions of how to regulate an increasingly large, diverse number of actors. Regulating space under international law has always presented challenges. The Outer Space Treaty, ratified by more than one hundred countries, is the only international legal framework focused on the exploration of space for peaceful purposes [167], and bans national appropriation of celestial objects such as the Moon, although the agreement has not been updated for decades and is vague on its stipulations [168]. The U.S. and a handful of other countries have also signed the Artemis Accords, an agreement intended to facilitate collaboration among partners in the pursuit of space exploration [167]. But some observers have raised the concern that "outer space could turn into the 'Wild West' of the twenty-first century" because of the combination of ambiguity and outdatedness of the space regime and the growing number of actors in space [169]. There are ongoing efforts to address this concern, with the U.S. recently announcing its intent to reform outdated regulations [170]. However, the specific format of this regulation, as well as how compatible it and any foreign attempts at reformation or legislation will be, remains unclear.

Scholars have devoted considerable attention to the number of actors as catalysts and constraints for competition and cooperation alike [171]. Attempts at cooperation between states will require agreements that are viable both at the international level, meeting the concerns of all state parties, and the domestic level, including the New Space actors empowered by the current wave of partnerships [172]. As commercial partners become critical infrastructure, they will likely demand their home state's protection and support of their interests at the international

level [173]. States may even be motivated to allow these interests to strategically constrain their bargaining with international partners, demanding that they accommodate their domestic commercial interests to make any international agreement viable at home. Thus, the rise of New Space may restrict the opportunities for necessary agreements as much as they make such accords necessary.

New Space's integration further complicates governance by demonstrating that there is no longer a clear distinction between the increasingly numerous commercial and government actors. The willingness of governments has led them to incorporate commercial space actors to the degree where their activities have converged. As private partners are "endowed" with "public responsibilities," scholars worry that the dynamic distorts state authority and obfuscates private liability, and thus promotes mutual accusations of blame during crises [173]. The 1972 Space Liability Convention showcases the danger of allowing New Space's growth to outpace governance: should the vast expansion of its activities threaten infrastructure, it is not only companies that are held responsible, but their host states as well. If New Space actors are continuously empowered financially and socially, their standing may aid them in attempts to avoid such blame, further reducing their accountability. In essence, public-private partnerships may create a moral hazard for New Space, where the backing of the government gives the sector confidence to experiment and test technical or legal boundaries, knowing that their patron affiliations will cushion consequences.

The distortion of responsibilities is also made worse by the inherent dual-use nature of commercial space infrastructure. In addition to reducing transparency, as commercial services at times fall under the umbrella of confidential national security [173], they also become likely targets for warfighting strategies [174], or simply collateral damage thereof [175]. Despite claims that the value and proliferation of these services internationally will, by nature, reduce the chances of conflict, (inter)national space law must account for ambiguity and draw distinctions in the event that the salve of economic interests proves insufficient [85]. Ideally, attempts to address the dual use dilemma in space would utilize lessons from arms races and control, restricting the militarization of infrastructure and limiting the proliferation of dual-use technologies across states. However, while no New Space member would wish to be a target of military action, the profit opportunity to produce dual-use assets for deterrence and arms races may reduce the sector's willingness to press for governance of space militarization. The efficiency afforded by New Space, and geopolitical demand for its services have made a major contribution to militarization inevitable. New global regulations are necessary to prevent runaway escalation of more overtly militarized partnerships that threaten to outpace the peace-making effects of more civil applications.

Further, in a key example for the need of such agreements where they become more complex with the rise of New Space, space traffic coordination requires multiple actors to work together to avoid collisions. This is especially true in L.E.O., as New Space activities continue to fill the orbits with satellite constellations and debris from payloads. As Oltrogge and Christensen note, the generalized trend of commercializing space is likely to result in an increase in space debris and, relatedly, a twenty to fifty-fold increase in space traffic management demands [37]. OneWeb and SpaceX began launching proliferated communications satellite constellations into L.E.O. in 2019, and demand for data, plus the decreasing cost, makes commercial satellites all the more attractive [176]. However, in 2022, the World Economic Forum released a statement of concern about the risks of commercializing space [177]. They note that about 11,000 satellites already orbit the Earth and debris is already a problem that threatens the sustainability of other satellites, particularly through a danger of Kessler Syndrome, in which orbital collisions can produce a cloud of debris that risks new collisions. New Space contributes to this danger, but currently plays little role in ameliorating the issue of coordination at the international level.

The current lack of international provision for regulating the issue of space debris can only increase as a concern in proportion with the

growth of private space activities. The enforcement problem is serious, as the responsibility to clean up both remnant and newly-added debris is fundamentally a collective action problem for space powers old and new [178]. As the number of actors continues to grow, the importance of a global governance framework markedly increases.

In 2022, the White House Office of Science and Technology Policy proposed becoming a focal point to coordinate cross-government action, a first step toward addressing the problem of orbital debris. But thus far, no meaningful progress has been made, despite the magnitude of the issue increasing every year [179]. New Space can make international action on space debris easier, even as the sector's growing activities simultaneously contribute to the challenge. The proliferation of debris, importance of sustainability, and increased danger of collisions all point to incoming market demand for debris removal. If entrepreneurial attempts at cleanup are as successful and effectively integrated into government strategy as launch services have been, the countervailing interests may allow New Space to solve the challenge of space debris where government efforts have been, if not absent, then continually sluggish.

8. Internationalizing the new space model

The amount of government investment in private space firms has increased considerably over the last two decades, catalyzing the growth of the U.S. private space sector. Correspondingly, the frequency of private space launches has increased, as has the activation of vast constellations of smallsats that have both commercial and security applications [180,181]. Estimates suggest that the global space industry will be worth more than \$1 trillion by 2040, a 300% increase from the current value, with a commensurate increase in the number of satellites in space, launches, and exploration [176]. The initial success of New Space in the U.S. has led other countries to promote their own commercial space sectors, emulating parts, but not the entirety of the U.S. approach. These differences are contingent on how states diverge from the willingness and opportunity factors to motivate the initial adoption. For willingness, most states lack the geopolitical rivalries that make the need for an independent space sector apparent, and the few that do now face major challenges in manifesting such private partners on demand. For opportunity, only the U.S. saw a major and productive New Space sector emerge organically, in large part due to the presence of private sponsors and venture capital. Lacking the same environment, other states have had to directly sponsor the emergence of commercial space companies, often in approaches that distort the benefits of the U.S. experience.

Members of the European Space Agency were early emulators of the U.S. approach, spurred by its early successes to attempt to commercialize many of the same services [3]. Under the umbrella of "Space 4.0," Europe has sought to integrate its public and private sectors [182], levy ambitious exploration and infrastructural objectives [4], and back these efforts with direct funding from ESA. However, U.S. New Space leveraged a broad range of funding sources to reach the technological niche that makes them efficient in the hyper-specialized commercial landscape.

H.E. McCurdy has distinguished indirect government sponsorship, entrepreneur patronage, and independent awards as the archetypal alternatives to traditional contracting [43]. The rise of entrepreneurs in Europe, such as Richard Branson, has made self-funding a real possibility for a few companies, but the overall difference is investment culture outside the U.S., which has been slow to adopt "venture capital" methods [62]. Likewise, the funding format of the European Space Agency invites parochial and particularist tendencies in funding disbursement [183]. To overcome these challenges, government sponsorship remains a key tool, but also helps explain why New Space has taken longer to expand in Europe, particularly as Old Space company Airbus/Thales Alenia continues to assume a dominant role in the commercial market there.

Russia, in a mirror to the American realization of deteriorating cooperation in space, likewise sought to promote a private space sector. However, issues of corruption have obstructed these efforts, and multiple government interventions have repeatedly restructured the commercial space sector. This has resulted in an inconsistent environment for investment, and one ill-prepared to meet the government's ambitious military and status-enhancing goals. As the war with Ukraine draws resources and personnel, while prompting the flight of experienced technical experts, Russia still faces major challenges and multiple strong competitors with larger economies and growing leads in international market share.

China has also moved away from being a fully-monopolized state industry and towards a more competitive, commercialized space sector. Russia and China have increasingly collaborated on joint space research and in 2021 announced the joint construction of the International Lunar Research Station [184], which has begun expanding membership with South Africa, Azerbaijan, and Pakistan in 2023 [185–187]. However, its aims of landing astronauts on the moon before 2030 necessitates a multipronged strategy that includes private industry that can support state goals [188]. The National Bureau of Asian Research estimated in 2021 that China has about 120–150 commercial space companies; commercial in the sense that they are not entirely state owned and up to 50% privately funded [189]. Although China's version of commercial space is one with more government oversight and intervention than in the West, a Chinese State Council White Paper appeared to offer a more permissive definition of "commercial" compared to a 2016 paper that restricted these activities to international space-related activities such as satellite exports. China now appears to endorse a commercial approach to all of its space industries: "[t]o serve the economy and society, China has promoted public and commercial application of its satellites and space technology ... The commercial use of satellite technology is thriving, which expands the application market for governments, enterprises and individuals" [189]. The promotion of a commercial space sector demonstrates a common recognition of the strategic value of space resources, orbital activities, and emerging technologies [191].

On the other hand, China's commercial investment has followed a different model, not the billionaire entrepreneurship that drove the U.S. side of the New Space boom. Despite allowing some decentralization to provincial sponsorship and entrepreneurship, the government still expects private enterprises to serve a supporting role to the primary state-owned enterprise (SOE), where "while some collaborations have occurred, SOEs tend to dismiss most emerging space companies" [192]. Subordination may guard against the risk of deviation between private profit motives and government strategic goals, but implicit and explicit government intervention may also deny new actors some of the independence to experiment and innovate. Concurrently, the sector has vastly expanded in terms of technical capability [11], but without the foundation of a preceding legal framework to effectively govern these activities [190]. Efforts to address these challenges are underway [193], and thus may assist in matching the improving private funding environment to produce an internationally competitive sector [193,194]. However, with recent purges in the PLA Rocket Force, issues of corruption and government leadership remain open questions for a community still closely aligned with government sponsorship [195].

The Indian Space Research Organization (ISRO) has increasingly sought to leverage space to enhance its military capabilities and national prestige [7,196]. This includes, more recently, the advent of commercial actors, and New Space India Limited (NSIL) was established in 2019 to provide commercial products to its space agency but also foreign customers [197]. The motives again appear to be commercial and geopolitical, with one former military officer referring to space as an alternative to a petroleum-based economy and a way to establish global prominence and visibility [198]. However, India still faces some similar challenges to China, with the private sector still broadly subordinated to the national program, and its legal environment still in need of modernization to match technical capabilities [199]. Until private

efforts achieve a sustainable non-government market and rise to service more international customers, NSIL's fortunes may remain tied to its state sponsor.

Collectively, these efforts will greatly contribute to the growth of the international space economy. However, just as challenges have begun to manifest amongst the earliest adopters of New Space partnerships, we will likely see such issues arise, and potentially new concerns introduced. However, states can also observe with hindsight the experience of early adopters, and thus have the chance to preempt many of the same challenges with sufficient government attention. It remains to be seen whether such top-down regulation is forthcoming, as motives for New Space's integration still predominate in discourse, and recognition of challenges still lags until major focusing events open windows for concerted action.

9. Conclusion

Government partnerships with private industry are nothing new. However, the vast expansion of public-private partnerships, particularly with the inclusion of New Space companies, certainly is. Such private actors are increasingly critical to their countries' national space objectives and see integration into the uppermost echelon of civil and military projects. This policy comes at a time when the potential for those same companies to challenge both their home governments and an international governance regime built primarily for state actors is increasingly evident.

We argue and provide evidence that the turn to New Space is the result of a confluence of opportunities to utilize New Space capabilities and geopolitical demands that sufficiently raised the willingness of governments to partner with them. New Space firms launched first in the U.S. with a privately-funded, efficiently productive, and increasingly prestigious approach to space exploration. Simultaneously, the deterioration of U.S.-Russian cooperation and growing status competition with China forced the U.S. to break Old Space monopolies and embrace New Space offerings. This adoption has prompted additional states to attempt to emulate the success of these partnerships, though local conditions have modified the format of their investments and bounded the resemblance to the initial turn towards New Space.

The deepening and international proliferation of public-private partnerships has globalized not only opportunities but also the dangers engendered by the privatization boom. For all the prestige partnerships bring, their accomplishments have validated a brand of politically activist entrepreneurs. These figures sometimes diverge strongly from government interests and pursue particularistic objectives, making their financial and social empowerment an inescapable concern. Likewise, the depth of integration of commercial space into dual-use applications inevitably invites greater suspicion and risks contributing to trends in space militarization. All the while, the growth of private space activities outpace (inter)national legal frameworks. First-mover advantages on technical and market niches, as well as limited orbital real estate suggest that all these concerns come with a shrinking time limit.

Because the motives for New Space's integration are unlikely to fade in the near future, efforts to address the negative externalities must therefore concomitantly increase. While the story of commercialization has begun with the U.S. case, the journey has only begun and will not remain confined to the first who embrace the trend. Commercialization by New Space actors is creating a new landscape and the need to understand the public-private partnerships, politics, and strategy of these developments [200]. Future research should continue mapping the contours of the shift to new private actors, explore the evolution of individual state motives and methods for accommodating the change, and evaluate the global consequences.

CRedit authorship contribution statement

Avishai Melamed: Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Adi Rao:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Olaf de Rohan Willner:** Conceptualization, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Sarah Kreps:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Replication materials for figures are available at:

[Replication Data for: Going to Outer Space with New Space: The Rise and Consequences of Evolving Public-Private Partnerships \(Reference data\) \(Dataverse\)](#)

Acknowledgments

We thank R. Lincoln Hines, Christopher Way, and Amelia C. Arsenault for their comments on this project.

References

- [1] C. Krauthammer, Closing the new frontier 2010, *Natl. Rev.* (2010). <https://www.nationalreview.com/2010/02/closing-new-frontier-charles-krauthammer/>.
- [2] T. Pultarova, NASA Orders 3 More SpaceX Crew Dragon Flights to Space Station, *Space, com*, 2022.
- [3] S. Lieberman, Harald Athanopoulos Köpping, T. Hoerber, *The Commercialisation of Space: Politics, Economics and Ethics*, first ed., Routledge, 2023.
- [4] J. Woerner, *Moon Village: A Vision for Global Cooperation and Space 4.0*, European Space Agency, 2016.
- [5] C. Davenport, As private companies erode government's hold on space travel, NASA looks to open a new frontier, *Wash. Post* (2021).
- [6] P.W. Singer, *Corporate warriors: the rise of the privatized military industry and its ramifications for international security*, *Int. Secur.* 26 (2001) 186–220.
- [7] D. Stroiokos, China, India, and the social construction of technology in international society: the English School meets Science and Technology Studies, *Rev. Int. Stud.* 46 (2020) 713–731, <https://doi.org/10.1017/S0260210520000273>.
- [8] A. Geppert, Rethinking the space age: astroculture and technoscience, *Hist. Technol.* 28 (2012) 219–223, <https://doi.org/10.1080/07341512.2012.722789>.
- [9] L. Bromberg, *NASA and the Space Industry*, John Hopkins University Press, 2000.
- [10] R. Margetta, NASA selects companies to develop commercial destinations in space, *NASA* (2021). <http://www.nasa.gov/press-release/nasa-selects-companies-to-develop-commercial-destinations-in-space>. (Accessed 22 June 2022).
- [11] G. Denis, D. Alary, X. Pasco, N. Pisot, D. Texier, S. Toulza, From new space to big space: how commercial space dream is becoming a reality, *Acta Astronaut.* 166 (2020) 431–443, <https://doi.org/10.1016/j.actaastro.2019.08.031>.
- [12] K. Bousedra, Downstream space activities in the new space era: paradigm shift and evaluation challenges, *Space Pol.* 64 (2023).
- [13] S. Chatlani, *NewSpace: an Era of Entrepreneurial Branding*, Georgetown University, 2018.
- [14] D. Paikowsky, What is new space? The changing ecosystem of global space activity, *New Space* 5 (2017) 84–88, <https://doi.org/10.1089/space.2016.0027>.
- [15] J. Klein, Rethinking requirements and risk in the new space age. <https://www.cnas.org/publications/reports/rethinking-requirements-and-risk-in-the-new-space-age>, 2019. (Accessed 22 June 2022).
- [16] J. Foust, Masten space systems files for bankruptcy. *SpaceNews* 2022. <https://spacenews.com/masten-space-systems-files-for-bankruptcy/>.
- [17] E. Berger, Blue Origin's powerful BE-4 engine is more than four years late—here's why, *Ars Technica* (2021). <https://arstechnica.com/science/2021/08/blue-origins-powerful-be-4-engine-is-more-than-four-years-late-heres-why/>. (Accessed 22 June 2022).
- [18] V. Kim, R. Pérez-Peñ, A.E. Kramer, Elon Musk refused to enable Ukraine drone attack on Russian fleet, *N. Y. Times* (2023).
- [19] S. Erwin, SpaceX launches NROL-85 spy satellite for U.S. National reconnaissance office. *SpaceNews* 2022. <https://spacenews.com/spacex-launches-nrol-85-spy-satellite-for-u-s-national-reconnaissance-office/>. (Accessed 22 June 2022).

- [20] P.L. Portanova, Acquiring launch services for the national reconnaissance office in the evolved expendable launch vehicle ERA. 54th International Astronautical Congress of the International Astronautical Federation, the International Academy of Astronautics, and the International Institute of Space Law, American Institute of Aeronautics and Astronautics, 2012, <https://doi.org/10.2514/6.IAC-03-V.2.03>.
- [21] D. Strokos, *International Relations and Outer Space*, Oxford Research Encyclopedia of International Studies, 2022.
- [22] D.-J. Jo, E. Gartzke, Determinants of nuclear weapons proliferation, *J. Conflict Resolut.* 51 (2007) 167–194.
- [23] B. Most, H. Starr, *Inquiry, Logic, and International Politics: with a New Preface by Harvey Starr*, University of South Carolina Press, 2015.
- [24] E. Gartzke, Kant we all just get along? Opportunity, willingness, and the origins of the democratic peace, *Am. J. Polit. Sci.* 42 (1998) 1–27, <https://doi.org/10.2307/2991745>.
- [25] E. Berger, The Large Vulcan Rocket Is Unlikely to Make its Debut in 2022, *Ars Technica*, 2022. <https://arstechnica.com/science/2022/06/the-large-vulcan-rocket-is-unlikely-to-make-its-debut-in-2022/>. (Accessed 22 June 2022).
- [26] H.E. McCurdy, Boeing: pairing commercial work with government contracts to reduce risk, in: *Financing the New Space Industry*. Financing the New Space Industry, Springer, 2019, pp. 37–44.
- [27] W. Harwood, Russia Severs Ties with U.S. And European Space Projects; ISS Operating Normally for Now, *CBS News*, 2022. <https://www.cbsnews.com/news/russia-severs-ties-with-u-s-european-space-projects-iss-operating-normally-for-now/>. (Accessed 22 June 2022).
- [28] E. Helmore, 'We're in a space race': nasa sounds alarm at Chinese designs on moon, *The Guardian* (2023).
- [29] A. Steinberg, Space policy responsiveness: the relationship between public opinion and NASA funding, *Space Pol.* 27 (2011) 240–246.
- [30] C. Davenport, The Space Barons: Elon Musk, Jeff Bezos, and the Quest to Colonize the Cosmos, *Public Affairs*, 2018.
- [31] E. Berger, Liftoff: Elon Musk and the Desperate Early Days that Launched SpaceX, Harper Collins, 2021.
- [32] A.D. Sorkin, The race to leave planet earth, *The New Yorker*. *The New Yorker* (2021). <https://www.newyorker.com/magazine/2021/07/26/the-race-to-leave-planet-earth>. (Accessed 22 June 2022).
- [33] S. Rementeria, Power dynamics in the age of space commercialisation, *Space Pol.* 60 (2022) 101472, <https://doi.org/10.1016/j.spacepol.2021.101472>.
- [34] P. Frankowski, *Outer space and private companies: consequences for global security*, *Routledge Handbook of War, Law and Technology* (2019) 280–292.
- [35] B. Dobos, Tortoise the titan: private entities as geoeconomic tools in outer space, *Space Pol.* 60 (2022) 101487, <https://doi.org/10.1016/j.spacepol.2022.101487>.
- [36] L. Heracleous, D. Terrier, S. Gonzalez, NASA's capability evolution toward commercial space, *Space Pol.* 50 (2019) 101330, <https://doi.org/10.1016/j.spacepol.2019.07.004>.
- [37] D. Oltrogge, I. Christensen, Space governance in the new space era, *Journal of Space Safety Engineering* 7 (2020) 432–438, <https://doi.org/10.1016/j.jsse.2020.06.003>.
- [38] National Coordination Office for Space-Based Positioning, Navigation, and Timing, *National Space Policy* (2020). <https://www.space.commerce.gov/policy/national-space-policy/>.
- [39] National Coordination Office for Space-Based Positioning, Navigation, and Timing, *National Space Policy* (2010). <https://www.gps.gov/policy/docs/2010/>.
- [40] C. Impey, *SpaceX and the Science of Failure*, The Hill, 2023.
- [41] R.D. Launius, Historical analogs for the stimulation of space commerce, *NASA, Office of Communications, Public Outreach Division, History Program Office* (2014).
- [42] S. Gonzalez, The astropreneurial Co-creation of the new space economy, *Space Pol.* 64 (2023), <https://doi.org/10.1016/j.spacepol.2023.101552>.
- [43] H.E. McCurdy, *Financing the New Space Industry: Breaking Free of Gravity and Government Support*, Springer Nature, 2019.
- [44] D. Messier, Soaring investment in commercial space dominated by handful of companies – parabolic arc, *Parabolic Arc* (2020). <http://www.parabolicarc.com/2020/03/12/soaring-investment-in-commercial-space-dominated-by-handful-of-companies/>. (Accessed 26 June 2022).
- [45] *USAspending.gov*, Recipient Profile: Space Exploration Technologies Corp, 2023.
- [46] NAIS, NAIS (2019). <https://prod.nais.nasa.gov/cgi-bin/nais/welcome.cgi>. (Accessed 26 June 2022).
- [47] R.M. Siverson, H. Starr, Opportunity, willingness, and the diffusion of war, *Am. Polit. Sci. Rev.* 84 (1990) 47–67, <https://doi.org/10.2307/1963629>.
- [48] S. Singh, C.R. Way, The correlates of nuclear proliferation: a quantitative test, *J. Conflict Resolut.* 48 (2004) 859–885.
- [49] B.R. Early, Exploring the final frontier: an empirical analysis of global civil space proliferation, *Int. Stud. Q.* 58 (2014) 55–67.
- [50] A. Thomas, The early days of South Korean rocket technology. *The Commercialisation of Space*, first ed., Routledge, 2023, p. 19.
- [51] D. Oberhaus, How did Jeff Bezos's Blue Origin fail to dominate the billionaire space race? *The Guardian* (2021).
- [52] E. Musk, Risky business: why Mars is more important than cosmetics and why a failed launch is also a partial success, *IEEE Spectrum* (2009). <https://spectrum.ieee.org/elon-musk-spacex>.
- [53] C. Anderson, Elon musk's mission to Mars, *Wired Magazine* (2012). <https://www.wired.com/2012/10/ff-elon-musk-qa/>.
- [54] A. Vance, Elon Musk's Space Dream Almost Killed Tesla, *Bloomberg*, 2015. <https://www.bloomberg.com/graphics/2015-elon-musk-spacex/>.
- [55] B. Bjelde, I am SpaceX employee #14, aerospace engineer, and VP of Human Resources, Ask me anything! *Reddit* (2016). https://www.reddit.com/r/spacex/comments/4vt74k/i_am_spacex_employee_14_aerospace_engineer_and_vp/.
- [56] National Academies of Sciences, *Issues Affecting the Future of the U.S. Space Science and Engineering Workforce: Interim Report*, 2006.
- [57] NASA, Chapter 5: NASA Manpower Policy, 2014.
- [58] Erickson, How Many Contractors Work for Nasa? *eclipseaviation.Com*, 2022. <https://www.eclipseaviation.com/how-many-contractors-work-for-nasa/>. (Accessed 26 June 2022).
- [59] The Associated Press, NASA Needs New Blood, Report Recommends, *NBC News*, 2006. <https://www.nbcnews.com/id/wbna12646982>. (Accessed 26 June 2022).
- [60] K. Cowing, NASA's Workforce Is Aging - NASA Watch, *NASA Watch*, 2018. <http://nasawatch.com/archives/2018/05/nasas-workforce.html>. (Accessed 26 June 2022).
- [61] J.C. Moltz, The changing dynamics of twenty-first-century space power, *J. Strat. Secur.* 12 (2019) 15–43.
- [62] S. Paladini, *The new frontiers of space: economic implications. Security Issues and Evolving Scenarios*, first ed., Palgrave Macmillan Cham, 2019.
- [63] H.E. McCurdy, Other methods of government support. *Financing the New Space Industry*, first ed., Palgrave Pivot, Cham, 2019.
- [64] A. Hein, Rosete C. Bruce, Space-as-a-Service: a framework and taxonomy of-as-a-Service concepts for space, in: *Proceedings of the International Astronautical Congress*, 2022.
- [65] E. Berger, NASA chief says cost-plus contracts are a "plague" on the space agency, *Ars Technica* (2022). <https://arstechnica.com/science/2022/05/nasa-chief-says-cost-plus-contracts-are-a-plague-on-the-space-agency/>. (Accessed 26 June 2022).
- [66] D.K. McQuiggan, R.J. Birk, The aerospace corporation. *Assessing Commercial Solutions for Government Space Missions*, Center for Space Policy and Strategy, 2022; February 2022.
- [67] Gallup, 50 Years After Moon Landing, Support for Space Program High, 2019.
- [68] D. Benton, Supply and demand in the commercial space-launch marketplace. 23rd Joint Propulsion Conference, 1987.
- [69] M. Forrest, *National Security Space Launch Report*, vol. 503, Rand Corporation, 2006.
- [70] S. Erwin, EELV is no more. It is now 'National Security Space Launch.', *Space News* (2019). <https://spacenews.com/eelv-is-no-more-it-is-now-national-security-space-launch/>. (Accessed 26 June 2022).
- [71] Vandenberg Space Force Base, Evolved Expendable Launch Vehicle (EELV), Vandenberg Space Force Base, 2021. <https://www.vandenberg.spaceforce.mil/About-Us/Fact-Sheets/Display/Article/1266632/evolved-expendable-launch-vehicle-eelv/>. (Accessed 26 June 2022).
- [72] A. Ansar, B. Flyvberg, How to solve big problems: bespoke versus platform strategies, *Oxf. Rev. Econ. Pol.* 38 (2022) 338–368.
- [73] T.A. Heppenheimer, *The Space Shuttle decision: NASA's search for a reusable space vehicle*. National Aeronautics and Space Administration, NASA History Office, Office of Policy and Plans, 1999.
- [74] R.D. Launius, *Frontiers of Space Exploration*, Greenwood Press, 2004.
- [75] P. Duggins, *Final Countdown: NASA and the End of the Space Shuttle Program*, UP Press, 2009.
- [76] R.D. Launius, Assessing the legacy of the space Shuttle, *Space Pol.* 22 (2006) 226–234.
- [77] L. Farrar, Experts: reliance on Russia makes NASA weak, *CNN.Com* (2008). <https://edition.cnn.com/2008/TECH/space/08/14/nasa.russia.soyuz/>. (Accessed 26 June 2022).
- [78] D. Larson, T. Paul, W. Wohlforth, *Status in World Politics*, Cambridge University Press, 2014.
- [79] D.C. Kang, Hierarchy and legitimacy in international systems: the tribute system in early modern East Asia, *Secur. Stud.* 19 (2010) 591–622, <https://doi.org/10.1080/09636412.2010.524079>.
- [80] D.W. Larson, A. Shevchenko, Status seekers: Chinese and Russian responses to US primacy, *Int. Secur.* 34 (2010) 63–95, <https://doi.org/10.1162/isec.2010.34.4.63>.
- [81] H. Morgenthau, *Politics Among Nations*, Revised, Knopf, New York, 1978.
- [82] J. Renshon, *Fighting for Status: Hierarchy and Conflict in World Politics*, Princeton University Press, Princeton, NJ, 2017.
- [83] P. Røren, Status seeking in the friendly Nordic neighborhood. *Cooperation and Conflict, Cooperat. Conflic* 54 (2019) 562–579.
- [84] M. Sheehan, *The International Politics of Space*, first ed., Routledge, London, 2007.
- [85] W. Cobb, *Privatizing Peace: How Commerce Can Reduce Conflict in Space*, Taylor & Francis, 2020.
- [86] M. Murray, *The Struggle for Recognition in International Relations: Status, Revisionism, and Rising Powers*, Oxford University Press, New York, 2018.
- [87] L. Gilady, *The Price of Prestige: Conspicuous Consumption in International Relations*, University of Chicago Press, Chicago, 2018.
- [88] D. Paikowsky, *The Power of the Space Club*, Cambridge University Press, 2017.
- [89] B. Kennedy, A. Tyson, *Americans' Views of Space: U.S. Role, NASA Priorities and Impact of Private Companies*, 2023.
- [90] M.E. Carranza, Rising regional powers and international relations theories: comparing Brazil and India's foreign security policies and their search for great-power status, *Foreign Pol. Anal.* 132 (2017) 255–277.
- [91] N. Leveringhaus, K.S. de Estrada, Between conformity and innovation: China's and India's quest for status as responsible nuclear powers, *Rev. Int. Stud.* 44 (2018) 1–22.

- [92] P. Musgrave, Defending hierarchy from the moon to the Indian ocean: symbolic capital and political dominance in early modern China and the Cold war, *Int. Organ.* 72 (2018) 1–36.
- [93] L. Nardon, Developed space programmes." the politics of space. A Survey, Routledge, 2011, pp. 58–76.
- [94] E. Seedhouse, The rise of SpaceX, in: E. Seedhouse (Ed.), *SpaceX: Starship to Mars – the First 20 Years*, Springer International Publishing, Cham, 2022, pp. 189–196, https://doi.org/10.1007/978-3-030-99181-4_10.
- [95] S. Ben-Itzhak, Space Blocs: the future of international cooperation in space is splitting along lines of power on Earth, *The Conversation* (2022).
- [96] SpaceRef Editor, Statement from Andrew Beal Regarding Cease of Operations by Beal Aerospace, 2000.
- [97] M. Von Bencke, *The Politics of Space: A History of US-Soviet/Russian Competition and Cooperation in Space*, Westview Press, 1997.
- [98] R.O. Keohane, J. Nye, Soft power and interdependence, *Survival* 15 (1973) 158–165.
- [99] H. Farrell, A.L. Newman, Weaponized interdependence: how global economic networks shape state coercion, *Int. Secur.* 44 (2019) 42–79.
- [100] D.W. Drezner, Bad debts: assessing China's financial influence in great power politics, *Int. Secur.* 34 (2009) 7–45.
- [101] D.C. Copeland, Economic interdependence and war: a theory of trade expectations, *Int. Secur.* 20 (1996) 5–41.
- [102] Cobb WN. Whitman, Commercialization and space: democracies can fly in space, *The International Journal of Space Politics & Policy* 19 (2021) 145–164, <https://doi.org/10.1080/14777622.2021.1988573>.
- [103] Apollo-Soyuz Test Project Overview, NASA, 2015. <http://www.nasa.gov/apollo-soyuz/overview>. (Accessed 26 June 2022).
- [104] R.J. Samuels, *Encyclopedia of United States National Security*, SAGE Publications, 2005.
- [105] K. Werth, A surrogate for war—the U.S. Space program in the 1960s, *Amerikastudien/American Studies* 49 (2004) 563–587.
- [106] J.C. Moltz, *The Politics of Space Security Strategic Restraint and the Pursuit of National Interests*, third ed., Stanford University Press, 2019.
- [107] F. Vidal, R. Privalov, A shrinking space power in the era of global change, *Space Pol.* 3 (2023), <https://doi.org/10.1016/j.spacepol.2023.101579>.
- [108] R.O. Keohane, *After Hegemony: Cooperation and Discord in the World Political Economy*, Princeton University Press, 2005.
- [109] M. Grinberg, Wartime commercial policy and trade between enemies, *Int. Secur.* 46 (2021) 9–52, https://doi.org/10.1162/isec_a_00412.
- [110] M. Hogenboom, America and Russia: uneasy partners in space, *BBC News* (2012).
- [111] W. Harwood, NASA uses final purchased Soyuz seat for Wednesday flight to station – spaceflight, *SpaceflightNow* (2020). <https://spaceflightnow.com/2020/10/13/nasa-uses-final-purchased-soyuz-seat-for-wednesday-flight-to-station/>. (Accessed 26 June 2022).
- [112] L. Cline, et al., Structuring future international cooperation: learning from the ISS. *Beyond the International Space Station: the Future of Human Spaceflight*, Springer, Dordrecht, 2002, pp. 39–54.
- [113] V. Zaborstkiy, Space engagement with Russia and Ukraine: preventing conflicts and proliferation, *Astropolitics* 4 (2006) 179–206.
- [114] J.C. Mauduit, Collaboration Around the International Space Station: Science for Diplomacy and its Implication for US-Russia and China Relations, *Secure World Foundation*, 2017.
- [115] K. Hammack, International relations in space: the role of miscalculation, militarization, and weaponization, *Astropolitics* 19 (2021) 230–236.
- [116] S. Erwin, Military launch poised for new era as Air Force prepares to move beyond EELV, *Space News* (2018). <https://spacenews.com/military-launch-poised-for-new-era-as-air-force-prepares-to-move-beyond-eelv/>. (Accessed 26 June 2022).
- [117] K. Hille, R. Wright, Russia Moves to Oust US from International Space Station, *Financial Times*, 2014.
- [118] Reuters, Russia Halts Deliveries of Rocket Engines to the U.S. *Reuters*, 2022.
- [119] R. Showstack, Congress examines NASA budget, space station, and relations with Russia, *Eos, Transactions American Geophysical Union* 95 (2014) 135–136, <https://doi.org/10.1002/2014EO160003>.
- [120] M. Daniels, *The History and Future of US China Competition and Co-operation in Space*, Defense Technical Information Center, November. 2020 Nov 1.
- [121] J.C. Moltz, The Russian space program: in search of a new business model, *Asia Policy* 27 (2020) 19–26.
- [122] S. Bugos, Russian ASAT test creates massive debris. *Arms control, Today Off.* 51 (2021) 30–31.
- [123] U.S. Erwin, Air Force sees no impact from Russia's decision to cut off supply of rocket engines, *Space News* (2022). <https://spacenews.com/u-s-air-force-sees-no-impact-from-russias-decision-to-cut-off-supply-of-rocket-engines/>. (Accessed 26 June 2022).
- [124] N. Larkin, Russia Will Quit International Space Station over Sanctions, *Bloomberg*, 2022.
- [125] T. Metcalfe, Russia will pull out of the International Space Station, space agency chief confirms, *Livest. Sci.* (2022). <https://www.livescience.com/russia-to-leave-international-space-station-chief-says>. (Accessed 26 June 2022).
- [126] A. Zemlianichenko, Trampoline to Space? Russian Official Tells Nasa to Take a Flying Leap, *NBC News*, 2014.
- [127] E. Palmer, Russia says NASA can use 'broomsticks' after stopping rocket engine sales, *Newsweek* (2022).
- [128] J. Maas, Does Britain have any choice but to compete in space? *The Commercialisation of Space: Politics, Economics and Ethics* (2023).
- [129] K. Chang, A. Troianovski, In space, U.S.-Russian cooperation finds a way forward, *N. Y. Times* (2022).
- [130] C. Davenport, SpaceX Launch Is Latest Sign of Lasting Russia-U.S. Space Partnership, 2022. *Washington Post*.
- [131] M. Sheehan, 'Did you see that, grandpa Mao?' The prestige and propaganda rationales of the Chinese space program, *Space Pol.* 29 (2013) 107–112, <https://doi.org/10.1016/j.spacepol.2013.03.003>.
- [132] R. Hines, Heavenly mandate: public opinion and China's space activities, *Space Pol.* 60 (2022), <https://doi.org/10.1016/j.spacepol.2021.101460>.
- [133] J. Weiss, J. Wallace, Domestic politics, China's rise, and the future of the liberal international order, *Int. Organ.* 75 (2021) 635–664. <http://doi:10.1017/S002081832000048X>.
- [134] R. Sutter, Positive equilibrium in U.S. - China relations: durable or not? A time of "testing" in Sino-American relations, *Maryland Series in Contemporary Asian Studies* (2009) 4, 2009.
- [135] J. Weiss, China and the future of world politics. *Perspectives on Politics, Perspect. Polit.* 15 (2017) 486–494, <https://doi.org/10.1017/S1537592717000160>.
- [136] D. Kliman, *Fateful Transitions: How Democracies Manage Rising Powers, from the Eve of World War I to China's Ascendancy*, University of Pennsylvania Press, 2014.
- [137] S. Chan, *Rumbles of Thunder: Power Shifts and the Danger of Sino-American War*, Columbia University press, 2022.
- [138] A. Wichowsky, J. Weiss, Getting tough on China: are campaign ads a signal of future policy or just cheap talk? *Legis. Stud. Q.* 46 (2021) 637–652.
- [139] J. Patton, J. Barr, Atlas and Delta capabilities to launch crew to low earth orbit. *AIAA SPACE 2009 Conference & Exposition, American Institute of Aeronautics and Astronautics*, Pasadena, California, 2009, <https://doi.org/10.2514/6.2009-6729>.
- [140] R.D. Launius, United States space cooperation and competition: historical reflections, *Astropolitics* 7 (2009) 89–100.
- [141] A. Chatzky, A. Siripurapu, S.J. Markovich, *Space Exploration and U.S. Competitiveness*, Council on Foreign Relations, 2021. <https://www.cfr.org/backgrounder/space-exploration-and-us-competitiveness>. (Accessed 27 June 2022).
- [142] S. Chan, *Looking for Balance: China, the United States, and Power Balancing in East Asia*, Stanford University Press, 2013.
- [143] Power transition, rising China, and the regime for outer space in a US-hegemonic space order, *Power Transition in the Anarchical Society: Rising Powers, Institutional Change and the New World Order*, Palgrave Macmillan Cham, 2022, pp. 329–352.
- [144] A. Johnston, China in a world of orders: rethinking compliance and challenge in Beijing's international relations, *Int. Secur.* 44 (2019) 9–60.
- [145] N. Adilov, P. Alexander, B. Cunningham, An economic "Kessler Syndrome": a dynamic model of earth orbit debris, *Econ. Lett.* 166 (2018) 79–82.
- [146] M. Elvis, T. Milligan, A. Krolikowski, The peaks of eternal light: a near-term property issue on the moon, *Space Pol.* 38 (2016) 30–38.
- [147] G. Allison, E. Schmidt, Is China Beating the US to AI Supremacy? *Belfer Center for Science and International Affairs, Harvard Kennedy School*, 2020 Aug.
- [148] K.A. Bingen, J. Kaitlyn, S.M. Zhanna, Russia threatens to target commercial satellites, *Center for Strategic & International Studies* (2022 Nov 10).
- [149] S. Erwin, Report: Industry Has to Face Reality that Commercial Satellites Will Be Targets in War, *Space News*, 2022.
- [150] A.A. Blanc, N. Beauchamp-Mustafaga, K. Holynska, M.S. Bond, S.J. Flanagan, Chinese and Russian Perceptions of and Responses to US Military Activities in the Space Domain, *RAND*, 2022.
- [151] A. Rogosa, Shifting spaces: the success of the SpaceX lawsuit and the danger of single-source contracts in America's space program, *Fed. Cir. BJ.* 25 (2015) 101.
- [152] J. Roulette, Blue Origin loses legal fight over SpaceX's NASA moon contract, *N. Y. Times* (2021).
- [153] C. Davenport, House Republicans Cite Concerns over SpaceX Explosion, *Washington Post*, 2016.
- [154] A. MacDonald, *The Long Space Age: the Economic Origins of Space Exploration from Colonial America to the Cold War*, Yale University Press, 2017.
- [155] Lockheed Martin Corp, Shareholders - CNN Business, 2023. <https://money.cnn.com/quote/shareholders/shareholders.html>.
- [156] Reuters, Elon Musk borrowed \$1 billion from SpaceX in same month of Twitter deal, *Wall Street Journal reports* (2023). *Reuters*.
- [157] D. Ljunggren, Anti-vaccine Canada truckers roll toward Ottawa, praised by Tesla's Musk, *Reuters* (2022).
- [158] V. Kim, Elon Musk acknowledges withholding satellite service to thwart Ukrainian attack, *N. Y. Times* (2023).
- [159] D. Deudney, *Dark Skies: Space Expansionism, Planetary Geopolitics, and the Ends of Humanity*, Oxford University Press, USA, 2020.
- [160] M. Rubenstein, *The Dangerous Religion of the Corporate Space Race*, University of Chicago Press, 2022.
- [161] H.E. McCurdy, *Space and the American Imagination*, second ed., Johns Hopkins University Press, 2011.
- [162] S. Rementeria, The evolving sociotechnical futures of outer space, *Futures* 152 (2023 Sep 1) 103220.
- [163] M. Sheetz, SpaceX looks to raise \$1.7 billion in new funding, boosting its valuation to \$127 billion, *CNBC* (2022). <https://www.cnbc.com/2022/05/22/el-on-musks-spacex-looks-to-raise-1point7-billion-in-new-funding.html>. (Accessed 28 June 2022).
- [164] S. Cao, SpaceX's booming government business is a threat to NASA and military contractors, *Observer* (2020). <https://observer.com/2020/10/spacex-nasa-military-contracts-threaten-government-contractors-lockheed-martin/>. (Accessed 28 June 2022).

- [165] S. Erwin, DoD withdraws Defense Production Act small launch contract awards, Space News (2020). <https://spacenews.com/dod-withdraws-defense-product-act-small-launch-contract-awards/>. (Accessed 28 June 2022).
- [166] Federal Aviation Administration, Comment from Space Exploration Technologies, Corp., 2023. <https://www.regulations.gov/comment/FAA-2023-1656-0009>.
- [167] R. Deplano, The Artemis accords: evolution or revolution in international space law? *Int. Comp. Law Q.* 70 (2021) 799–819, <https://doi.org/10.1017/S0020589321000142>.
- [168] European Space Agency. About Space Law. European Space Agency n.d. https://www.esa.int/About_Us/ECSL_-_European_Centre_for_Space_Law/About_space_law (accessed June 28, 2022).
- [169] H.M. Babcock, The public trust doctrine, outer space, and the global commons: time to call home, *ET. Syracuse L. Rev.* 69 (2019) 191.
- [170] J. Foust, Harris says U.S. To update commercial space regulations, Space News (2022). <https://spacenews.com/harris-says-u-s-to-update-commercial-space-regulations/>.
- [171] A. Moravcsik, Taking preferences seriously: a liberal theory of international politics, *Int. Organ.* 51 (1997) 513–553.
- [172] R. Putnam, Diplomacy and domestic politics: the logic of two-level games, *Int. Organ.* 42 (1988) 427–460.
- [173] L. Newlove-Eriksson, J. Eriksson, Governance beyond the global: who controls the extraterrestrial? *Globalizations* 10 (2013) 277–292.
- [174] B. Bowen, War in Space: Strategy, Spacepower, Geopolitics, Edinburgh University Press, 2020.
- [175] R. Dickey, Commercial normentum: space security challenges, commercial actors, and norms of behavior, Center for Space Policy and Strategy (2023).
- [176] Wharton Staff, Why big business is making a giant leap into space. Knowledge at wharton. <https://knowledge.wharton.upenn.edu/article/commercial-space-economy/>, 2019 (accessed June 28, 2022).
- [177] D. Broom, As private satellites increase in number, what are the risks of the commercialization of space? *World Economic Forum* (2022). <https://www.weforum.org/agenda/2022/01/what-are-risks-commercial-exploitation-space/>. (Accessed 28 June 2022).
- [178] C. Muñoz-Patchen, Regulating the space commons: treating space debris as abandoned property in violation of the outer space treaty, *Chicago Journal of International Law*. *Chicago Journal of International Law* 19 (2018).
- [179] T. Hitchens, Industry chivies White House for “immediate” action to clean up space junk, *Breaking Defense* (2022). <https://breakingdefense.sites.breakingmedia.com/2022/01/industry-chivies-white-house-for-immediate-action-to-clean-up-space-junk/>. (Accessed 28 June 2022).
- [180] J. Grunert, Sanctions and satellites: the space industry after the Russo-Ukrainian war, *War on the Rocks* (2022).
- [181] T. Malik, Elon Musk says SpaceX focusing on cyber defense after Starlink signals jammed near Ukraine conflict areas, *SpaceCom* (2022).
- [182] ESA Ministerial Council, What is space 4.0?. https://www.esa.int/About_Us/Ministerial_Council_2016/What_is_space_4.0, 2016.
- [183] M. Nicholas, S. Lieberman, Public-private partnerships and foreign direct investment for space, in: *The Commercialisation of Space: Politics, Economics and Ethics*, first ed., 2023.
- [184] M. Bergmann, A. Lohsen, Understanding the Broader Transatlantic Security Implications of Greater Sino-Russian Military Alignment, *Center for Strategic and International Studies*, 2022.
- [185] A. Jones, South Africa joins China’s moon base project, *Space News* (2023).
- [186] A. Jones, Azerbaijan Signs up to China’s International Moon Base Project, *Space News*, 2023.
- [187] A. Jones, Pakistan becomes latest country to join China’s ILRS moon project, *Space News* (2023).
- [188] B. Curcio, Developments in China’s Commercial Space Sector, 2021.
- [189] National Bureau of Asian Research, Developments in China’s commercial space sector: interview with blaine curcio, *Center for Innovation, Trade, and Strategy* (2021).
- [190] M. Zhang, X. Yang, China’s emerging commercial space industry: current developments, legislative challenges, and regulatory solutions, *Acta Astronaut.* 202 (2023) 9–16.
- [191] N. Goswami, China in space: ambitions and possible conflict, *Strategic Studies Quarterly* 12 (2018) 74–97.
- [192] I. Liu, E. Linck, B. Lal, K. Crane, X. Han, T. Colvin, Evaluation of China’s Commercial Space Sector, 2019.
- [193] Y. Yuan, W. Peeters, Research viewpoint: rapid growth of the Chinese commercial space sector, *Astropolitics* 17 (2019) 191–207, <https://doi.org/10.1080/14777622.2019.1675430>.
- [194] A. Jones, Chinese Space Launch Startups Attract a Frenzy of Investment, *Space News*, 2022.
- [195] J. Palmer, China Replaces Top Rocket Force Commanders, *Foreign Policy*, 2023.
- [196] M. Aliberti, India in Space: between Utility and Geopolitics, Springer International Publishing, 2018.
- [197] N. Bhattacharjee, Liftoff: modi’s space push for India counts on private players, *Reuters* (2023).
- [198] A. Lele, India’s foray into the commercial space market, *The Space Review* (2021). <https://www.thespacereview.com/article/4132/1>. (Accessed 28 June 2022).
- [199] N. Nagendra, P. Basu, Demystifying space business in India and issues for the development of a globally competitive private space industry, *Space Pol.* 36 (2016) 1–11, <https://doi.org/10.1016/j.spacepol.2016.02.006>.
- [200] R.P. Rajagopalan, Managing New Actors in the Space Domain, *The Diplomat*, 2019. <https://thediplomat.com/2019/06/managing-new-actors-in-the-space-domain/>. (Accessed 28 June 2022).